Portfolio

Portfolio Value & Operations

Operating decisions across the estate: cost, service, productive use and asset value

Portfolio: Needs attention

3 simulated sites · 1 mini-grid / 1 cold room / 1 charging hub

Assessment scope & periods ↗

Operating-cost exposure →

$3.31–3.97

Kobo · 30 min avoidable generator time

Modelled + scenario · one-hour window

Service & stock risk →

2 sites

Cold-room excursion + interrupted charging

Measured events · financial loss unpriced

Productive-use opportunity →

18–20 kW

Kobo · headroom on 3 of 5 days

Modelled · demand validation remains

Asset life →

4 early warnings

Battery below reserve · overdue servicing · charger fault

Early warning · not a condition check

Verified value recovered →

Not established

Intervention outcomes need more evidence

Completed work does not establish savings

3 priority sites · **1** opportunity · **3** site evidence gaps · **3** accepted actions · **8** warnings not yet priced

Distinct consequence categories · no combined savings total

Where management should act first

Findings ranked by consequence and urgency, with evidence confidence alongside.

6 FINDINGS

All priorities | Cost | Risks | Opportunities

6 findings

SITE	FINDING / OPERATING MECHANISM	VALUE / CONSEQUENCE	PRIORITY	CONFIDENCE	ACTION / NEXT MOVE
GreenHarvest Cold-chain	**Backup response failed** F-118 · Automatic start failed; cooling interrupted · Cooling continuity · Stock kept saleable	**Product conditions exposed** Measured peak 8.2°C · loss unpriced 6 Oct · 19:00–20:00	High	High Observed sequence	**ACT-051** Witness an automatic-start test Open
Kobo Mini-grid	**Extended generator support** F-104 · 30 min candidate; charging trade-off remains · Supply cost per kWh	**$3.31–3.97** Modelled fuel + scenario maintenance 6 Oct · 14:20–15:20	High	Low ⓘ PCS capability gap	**ACT-048** Verify the policy trial Outcome pending
Depot EV charging	**Charging session interrupted** F-130 · Fault-aborted session; root cause unknown · Buses ready to run	**1 interrupted session** Measured events · revenue loss unpriced 6 Oct · 08:00–16:00	Medium	Medium ⓘ 1 other outcome unknown	**ACT-061** Investigate connector 2 Open
GreenHarvest Cold-chain	**Temperature excursion** F-119 · Chamber above 4°C after backup failed · Stock kept saleable	**25–35 min above policy** Measured peak 8.2°C · no spoilage determination 6 Oct · 19:00–20:00	Medium	High Stock and loading not recorded	**Review recommendation** Record stock and connect the door log Open
Kobo Mini-grid	**Battery output-limit report missing** F-108 · One 14:50 report absent from the battery controller · Evidence quality	**30 min unclassified** Evidence gap · limits F-104 6 Oct · 14:20–15:20	Medium	High Reporting cause unknown	**ACT-049** Restore battery-limit reporting Open
Kobo Mini-grid	**Productive-use headroom** F-109 · Recurring renewable capacity available · Productive electricity sales	**18–20 kW usable** Modelled headroom · $2.50 scenario sales Separate 5-day study · 3 eligible days	Medium	Medium ⓘ Demand not established	**Review recommendation** Evaluate a 10 kW productive load Recommendation

Early warnings · not priced yet

Risk — **Repeated maintenance without a result**
GreenHarvest · Cooling continuity · Backup generator visited 3 times in 60 days (14 Aug, 18 Sep, 6 Oct); the automatic start still failed

Early warning — **Battery below its reserve**
Kobo · Battery and generator life · 5 of the last 30 days · lowest 22%

Early warning — **Backup start test overdue**
GreenHarvest · Compressor life · Monthly test due 2 Oct, not done

Early warning — **Charger fault and overdue inspection**
Depot · Charger life · Connector 2 fault on 6 Oct · inspection overdue since 30 Sep

Open all Findings →

Priority and confidence answer different questions.

Outcomes across the estate

How each site is doing on what matters

	Kobo	GreenHarvest	Depot
Revenue	**Unknown** Productive electricity sales	**Not measured yet** Paid cold-room use	**Watch** Charger use
Operating cost	**Needs attention** Supply cost per kWh	**Not measured yet** Cost per cooling unit	**Not measured yet** Energy cost per km
Service and stock	**Unknown** Reliable service	**Needs attention** Cooling continuity	**Needs attention** Buses ready to run
Asset life	**Watch** Battery and generator life	**Watch** Compressor life	**Watch** Charger life
Return on assets	**Not measured yet** Site profitability	**Not measured yet** Paid cold-room use	**On track** Room before expansion
Evidence	**Watch** Evidence quality	**Watch** Evidence quality	**Watch** Evidence quality

Needs attention · Watch · On track · Unknown · Not measured yet

Evidence that would change a decision

Restore Kobo's missing PCS capability report to resolve two indeterminate dispatch intervals.

Inspect the constraint →

GROW USEFUL DEMAND

An opportunity beyond reducing cost

Scenario

10 kW × 1 h — Grain-mill evaluation at Kobo

→

$2.50 — Scenario gross sales per eligible window

Within 18–20 kW headroom · $0.25/kWh assumed tariff · before costs. Validate demand and service guardrails before committing.

Review opportunity Finding →

Inspect physical headroom →

CLOSE THE LOOP

What changed in this review

Inspect basis ↗

- **1 maintenance activity completed** — Outcome unverified
- **1 policy trial awaiting outcome evidence** — Guardrails incomplete
- **0 verified resolutions in these examples** — Value not yet recovered

Observed change → comparable evidence → verified outcome → eligible financial value

Business context: Depot delivered **120 kWh** in complete records; **$36.00** scenario tariff estimate. Inspect energy basis →

Other sites lack comparable full-window energy coverage.

Canonical basis: v6.9 8.1 / 8.8 / 8.9; Client Demo Recommendation Screen 1 / A1 · **Question:** Where should management act first, what is at stake, and what would prove improvement?
Preserved: Finding-owned ranked rows; Site / Finding / Action root changes; Action references confined to the Action column. Financials reuses site-owned consequences; unknowns and claim limits remain explicit.
Adapted: A1 business-value hierarchy above the canonical priority table. Three existing sites; objective matrix instead of ungrounded percentages; productive-use and verification visible together. Counts are secondary context.
Deferred: Six-site estate, monthly extrapolation, productive-use share, comparable energy roll-up, asset valuation, reconciliation residual and verified recovered money require their own supporting records. None is inferred from this hour.
Content owners: PortfolioSummary · FindingIndex · PortfolioConsequenceRollup · SiteAssessmentSummary · CapabilityReadiness · ActionSummary · VerificationSummary · **Prototype route:** #/portfolio

Sites ›

Sites

Find and compare sites across the operating estate

Directory | Add site · form | From template | Batch upload | Setup history

All sites | Mini-grid | Cold-chain | EV charging

Sites	Needs attention	Evidence limited
3	3	3
Stable site identities	Evidence-backed operational reasons	Window-specific claim limits

Site inventory

Mode is provenance. A row is a Site, never a simulation run.

SITE	TYPE	MODE	ASSESSMENT	TOP ISSUE	EVIDENCE	LAST ANALYSED
Kobo Mini-grid MG-DEMO-01	Mini-grid	Simulated	Needs attention	**Generator support**	**Limited** ↗	6 Oct, 16:01 EAT · latest assessment
GreenHarvest Cold Room CR-DEMO-01	Cold-chain	Simulated	Degraded	**Temperature excursion**	**Limited** ↗	6 Oct, 20:03 EAT · latest assessment
Depot Charging Hub EV-DEMO-01	EV charging	Simulated	Needs attention	**Interrupted session**	**Limited** ↗	6 Oct, 16:03 EAT · latest assessment

In setup

2 sites not yet active · they appear in the directory once activated

+ Form | From template | Batch upload

SITE	TYPE	CREATED BY	STATE	NEXT STEP	CREATED
Lakeside Mini-grid MG-LAKE-01	Mini-grid	Form	Draft	**Connect the gateway and map points →**	6 Oct 10:12
Westfield Depot EV-WEST-01	EV charging	Template · EV depot	Pending activation	**Map 12 of 16 points →**	4 Oct 15:40

ⓘ A site with no accepted evidence in the selected period is Unknown. Missing telemetry is not proof of healthy operation.

Canonical basis: v6.9 8.2 · **Question:** Which site should I open?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: SiteDirectorySummary · SiteAssessmentSummary · CapabilityReadiness · **Prototype route:** #/sites

Kobo Mini-grid

Simulated site · MG-DEMO-01

60 kWp PV · 200 kWh battery · 40 kW genset · residential, clinic & grain mill

Needs attention · Confidence: Low ⓘ · Assessed this hour: dispatch only · other objectives need more evidence · Inspect evidence ↗

Objective: serve residential, clinic and productive loads while protecting a 30% battery reserve.

Supply cost per kWh
Needs attention
30 min of generator time avoidable in this hour
Trial under way · result pending
View detail →

Reliable service
Unknown
Power delivered is measured; what customers needed is not
Watched during the generator trial
View detail →

Productive electricity sales
Unknown
18–20 kW of solar spare on 3 of 5 days · sales not metered
Opportunity found · not acted on yet
View detail →

Battery and generator life
Watch
Battery below its 30% reserve on 5 of 30 days · generator service due soon
Early warning · condition not checked
View detail →

Evidence quality
Watch
One battery report missing · 30 min unclassified
Reporting fix open
View detail →

Not measured yet
Site profitability, Cash collected
Needs site income, cost and billing records

What happened in the selected hour

Plain sequence from accepted readings · the dispatch view has the detail

Open Dispatch →

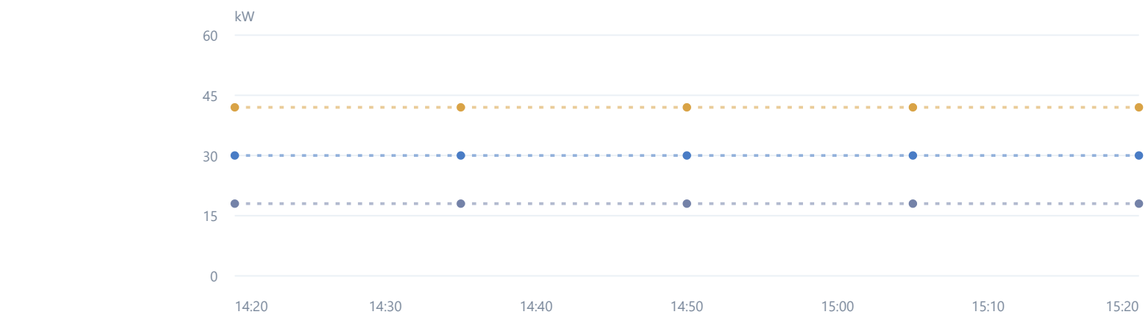
14:20	14:20	14:20–15:20	14:50	15:20
Generator running at 18 kW	**Sun and battery could carry the 30 kW load**	**Generator output kept charging the battery**	**Battery output-limit report missing**	**Generator still running**
Generator controller	42 kW solar available · battery 74%	Battery 74% → 88%	Battery controller	Stop setting is 90% battery charge

The generator kept running because its stop setting is 90% battery charge, not because the load needed it. Its output went into the battery.

Operating context

Selected hour · context for the cards, not a status source

Single line diagram →



Dots are readings every 15 minutes. Dotted lines join readings; they are not continuous measurement.

Needs attention

1 **Generator ran longer than needed**
Solar and battery could carry the load in two 15-minute blocks (F-104).
Performance › Dispatch →

2 **A battery report is missing**
The 14:50 battery output-limit report is absent, so 30 minutes stay unclear.
Evidence →

3 **Check the charging trade-off**
Generator energy charged the battery. Net savings are not yet shown.
Financials →

Bounded money consequence

$3.31–3.97

Cost exposure for the avoidable 30 minutes: modelled fuel plus a maintenance allowance. **Not realised savings.**

Open financial bridge →

Next step

Review the 90% generator stop setting

Trial a lower stop setting and judge it on generator time, service, battery reserve and stored energy together.

Review recommendation

Canonical basis: v6.9 8.18 · **Question:** Is the mini-grid meeting its operating objective?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Shared site-overview skeleton (all verticals): status with confidence → objective → canonical assessment cards → plain causal strip → context + needs attention (each with destination) → bounded consequence + next step. Only assessed objectives get a health conclusion; R enewable and Fuel use Unknown, never an opportunity or "not assessed" status.

Deferred: A site-wide Healthy label while service/asset objectives are unassessed.

Content owners: SiteOverview · DispatchIntervalRevision · OperationalTimeline · FinancialConsequence · **Prototype route:** #/mini

Kobo Mini-grid

Simulated site · MG-DEMO-01

Definition, topology and controls · public configuration at the time of evidence

Foundation v4 · Effective 6 Oct 2026 · 03:00 EAT · immutable historical reference

Review configuration readiness →

Configured topology

Ratings and configured connections; not live energy flows

Single line diagram →

PV array
60 kWp · 50 kW inverter

Battery + PCS
200 kWh · ±30 kW

Diesel generator
40 kW · 0.3 L/kWh

400 V AC BUS · registered electrical connections

Residential
18 kW example load

Clinic
4 kW · critical

Grain mill
8 kW · productive

Identity & operating obligations

Site identity	MG-DEMO-01
Archetype	Mini-grid
Timezone	Africa/Nairobi
Critical service	Clinic feeder
Productive use	Grain mill
Evidence provenance	Simulated

Financial assumptions are owned by Site Financials, separate from physical ratings.

Time-valid control configuration

CONTROL	CONFIGURED VALUE	AUTHORITY	EVIDENCE / CHANGE
Battery reserve	30% SOC	Foundation v4	Published policy ↗ ↗
Generator start / stop	35% / 90% SOC	Foundation v4	Review proposed change →
Minimum generator runtime	30 min	Foundation v4	Constraint evidence ↗ ↗
Source priority	PV → Battery → Generator	Foundation v4	See actual operation →

Canonical basis: v6.9 8.10 · **Question:** What is configured, and what must the site achieve?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: SiteFoundationVersion · SiteControls · TopologyReadModel · SignalBindingVersion · **Prototype route:** #/foundation

Kobo Mini-grid

Simulated site · MG-DEMO-01

Definition, commissioning and gateway readiness

Needs review — Configuration readiness is distinct from claim readiness

Definition	Topology	Specifications	Controls
Complete	Complete	Needs review	Conditional
Objective and critical loads named	Components and source bindings resolved	Battery capacity not field-verified	Published values available; change review open

Readiness and commissioning evidence

AREA	DECLARED	CONFIRMED	WHAT IS NEEDED	OWNER
Battery capacity	200 kWh nominal	Unverified	Controlled capacity test	Asset Health →
Generator fuel curve	Public 0.30 L/kWh	Not declared	Site fuel curve or a fuel-meter test	Raise action →
Device / signal mapping	Mapping v1	Configured	Commissioning witness record	Evidence →
Generator stop policy	90% SOC	Published	Review operational effect	Action →
Critical load obligation	Clinic prioritized	Limited	Requested versus delivered service evidence	Service →

Gateway status · GW-MG-01

How site data reaches AssetOps in the analysed window · 6 Oct 2026 · 14:20–15:20 EAT

Health › Data sources →

Status	Availability	Data completeness	Clock and delivery
Online	100%	95.0%	+0.4 s · synced
Last message 15:20:03 EAT · sent at 16:00 in one batch	0 outages · 0 min offline	19 of 20 readings · 1 late · largest gap 30 min · battery output limit (14:35 → 15:05)	Held on the gateway and sent in one batch (16:00 for the event hour)



Connected sources

Identity · Cellular 4G · −87 dBm · fair · firmware 2.4.1 · Point list v3 · applied 1 Oct 2026

- **Battery controller (PCS)** · Reporting · 1 missed report
- **Generator controller** · Reporting
- **Feeder meters (3)** · Reporting
- **Solar inverter** · Reporting
- **Fuel tank sensor** · Stale · opening reading only
- **Battery temperature** · Never reported

Single line diagram → Health › Data sources →

Effect on readiness and confidence

1 **Gateway stayed online all window**
The missing 14:50 report came from the battery controller, not the gateway.
Restore reporting · ACT-049 →

2 **Readings arrive in batches**
Conclusions for a window appear after the upload, not live.
Site evidence →

ⓘ This view assesses definitions, mappings, commissioning and the gateway. Claim ceilings and missing analytical inputs belong to Site Evidence.

Canonical basis: v6.9 8.10 · 8.35 · **Question:** Is the configuration sufficient and independently confirmed?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Restores configuration readiness without duplicating the Evidence capability table.
Deferred: General commissioning workflow and automatic field certification.
Content owners: FoundationReadiness · SiteFoundationVersion · CapabilityReadiness · **Prototype route:** #/readiness

Kobo Mini-grid

Simulated site · MG-DEMO-01

Configuration change CHG-019 · generator stop policy

CHG-019 · Proposed — Generator stop threshold · linked to ACT-048 — Open verification →

Proposed change	Target	Guardrails	Outcome
90% → 70%	≥25% lower	3 required	Not evaluated
Generator stop SOC threshold	Candidate generator runtime	Service, reserve and stored energy	A proposal is not applied evidence

Change and effect

FIELD	BEFORE	PROPOSED	PURPOSE
Stop threshold	90% SOC	70% SOC	Reduce extended generator support
Reserve floor	30% SOC	30% SOC	Protect reliability margin
Minimum runtime	30 min	30 min	Avoid short cycling
Service objective	Clinic priority	Clinic priority	Preserve critical service

Decision path

1 **Declare success before evaluating**
Freeze target, coverage, comparable windows and guardrails.
Inspect plan →

2 **Apply and observe**
The applied setting must be recorded, then later readings accepted.
Inspect accepted evidence →

3 **Retain, retune or revert**
Outcome is decided from target and guardrails together.
View verification →

ⓘ Illustrative proposal. No real controller command is sent by this mockup. A retune would create a new change record, preserving the prior one.

Canonical basis: v6.9 8.11 · **Question:** Did the configuration change improve the intended outcome?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: ConfigurationChange · VerificationPlan · VerificationOutcome · **Prototype route:** #/changes

Kobo Mini-grid

Simulated site · MG-DEMO-01

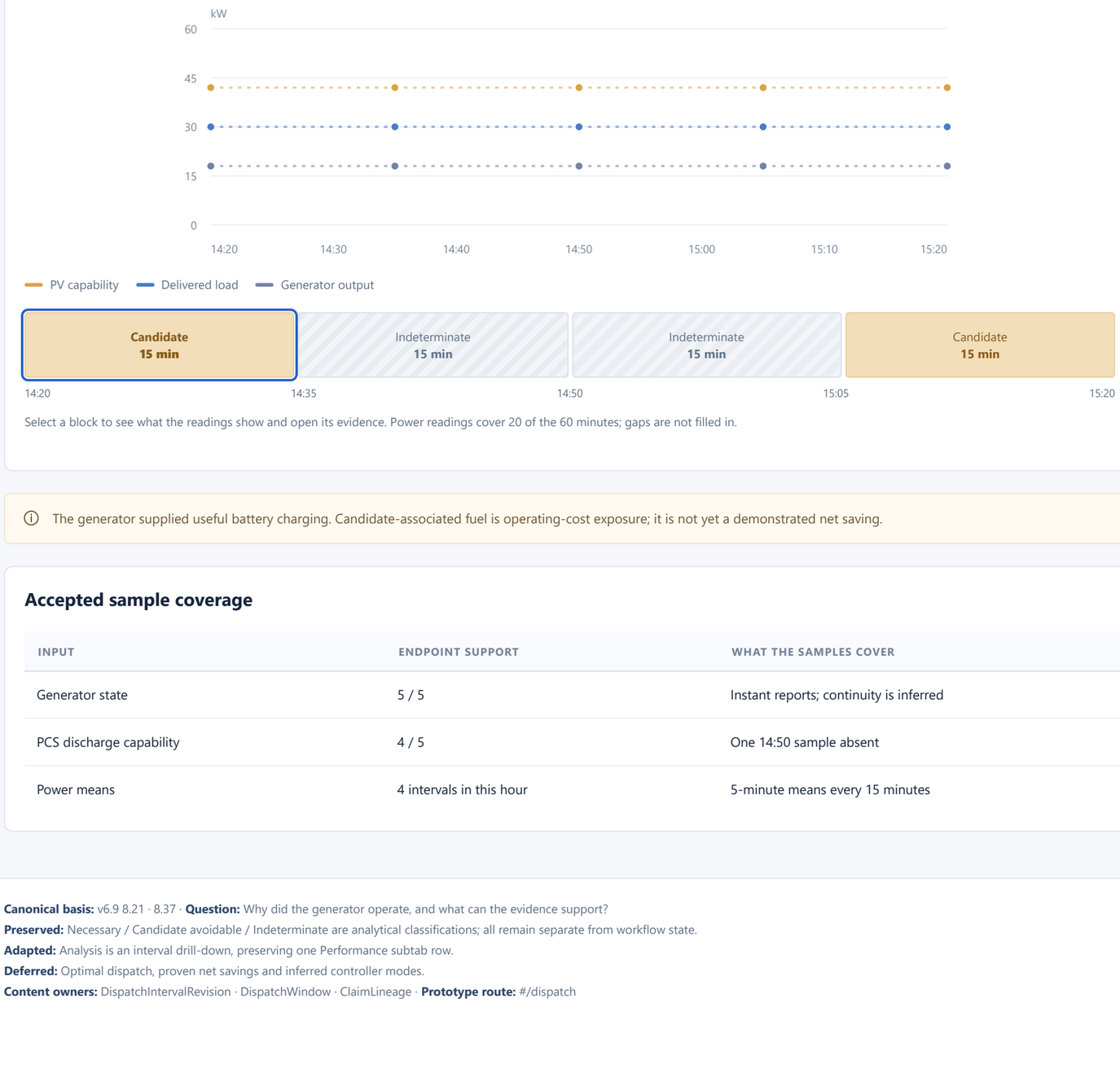
Performance · source coordination and generator support

Needs attention · Confidence: Low ⓘ Start and stop times uncertain by up to 25 min · one battery report missing — Inspect evidence ↗

Generator operation	Candidate avoidable	Necessary	Indeterminate
60 min	30 min	0 min	30 min
Inferred continuity from reported states	Supported portions under declared assumptions	Within this selected hour	PCS capability missing at 14:50

Sources and 15-minute blocks

Selected hour · 14:20–15:20 EAT



Select a block to see what the readings show and open its evidence. Power readings cover 20 of the 60 minutes; gaps are not filled in.

Selected block · 14:20–14:35

Select another block above

Candidate avoidable

Solar and battery could have carried the load; the generator was not needed for service.

Load served	30 kW OBSERVED
Solar available	42 kW OBSERVED
Battery charge	74 → 77.6% OBSERVED
Battery output limit	30 kW OBSERVED
Reserve floor	30% DECLARED
Generator output	18 kW OBSERVED
Operator override	None recorded OBSERVED

How this was worked out — Open F-104 →

ⓘ The generator supplied useful battery charging. Candidate-associated fuel is operating-cost exposure; it is not yet a demonstrated net saving.

Accepted sample coverage

View all evidence →

INPUT	ENDPOINT SUPPORT	WHAT THE SAMPLES COVER	CLAIM IMPACT
Generator state	5 / 5	Instant reports; continuity is inferred	Operating period reconstructed
PCS discharge capability	4 / 5	One 14:50 sample absent	Two adjacent segments indeterminate
Power means	4 intervals in this hour	5-minute means every 15 minutes	20 / 60 min measured; gaps not filled

Canonical basis: v6.9 8.21 · 8.37 · **Question:** Why did the generator operate, and what can the evidence support?
Preserved: Necessary / Candidate avoidable / Indeterminate are analytical classifications; all remain separate from workflow state.
Adapted: Analysis is an interval drill-down, preserving one Performance subtab row.
Deferred: Optimal dispatch, proven net savings and inferred controller modes.
Content owners: DispatchIntervalRevision · DispatchWindow · ClaimLineage · **Prototype route:** #/dispatch

Kobo Mini-grid

Simulated site · MG-DEMO-01

Performance · renewable use and productive-load opportunity

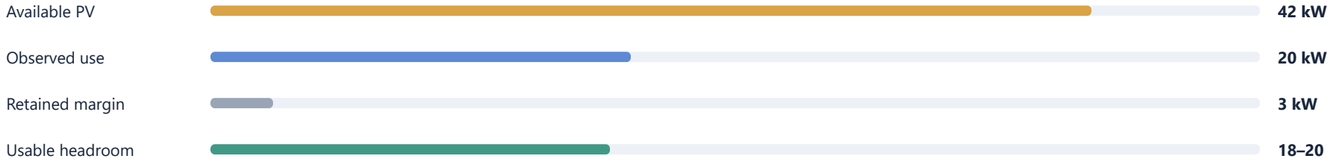
Watch | Confidence: Medium ⓘ | Lens: Opportunity | Solar left unused on 3 of 5 adequately observed days · separate 5-day study

Compare with Dispatch evidence →

PV capability	PV used	Usable headroom	Recurring opportunity
42 kW	20 kW	18–20 kW	3 / 5 days
Reported capability; ±1 kW model allowance	Measured output in eligible windows	After a 3 kW operating margin	One eligible hour per qualifying day

From physical headroom to an opportunity

Actual one-minute means cover each eligible one-hour window



[41, 43] kW capability bound − 20 kW load − 3 kW margin = 18–20 kW. No charging obligation or critical service deficit is evidenced in eligible windows.

Productive-use evaluation

Electrical feasibility does not establish customer demand

ILLUSTRATIVE GRAIN MILL

10 kW × 1 h

Up to 10 kWh of additional served energy. At a scenario tariff of $0.25/kWh: **$2.50 gross sales**, before other costs.

Inspect opportunity constraints ↗

Eligible and excluded windows

DAY / PERIOD	OPPORTUNITY	CONSTRAINT CHECK	DECISION
Day 1 · 12:00–13:00	18–20 kWh	Storage charge acceptance 0; service covered	Eligible
Day 2 · 12:00–13:00	—	Storage charging obligation	Excluded
Day 3 · 12:00–13:00	18–20 kWh	All required evidence present	Eligible
Day 4 · 12:00–13:00	—	Critical service evidence missing	Limited
Day 5 · 12:00–13:00	18–20 kWh	All required evidence present	Eligible

ⓘ Opportunity is an evidence-backed capacity estimate. A paired load-addition comparison and local demand validation are required before an investment or revenue claim.

Canonical basis: v6.9 8.23 · **Question:** Is recurring unused capacity practical for productive use?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Productive-use is a section within Renewable, not a third navigation level.
Deferred: Market demand forecasting and generalized scenario experimentation.
Content owners: RenewableOpportunityWindow · ServiceWindow · FinancialConsequence · **Prototype route:** #/renewable

Kobo Mini-grid

Simulated site · MG-DEMO-01

Evidence · capability readiness and claim ceilings

Evidence limited — 5 capabilities · 0 full · 3 limited · 2 none · no single score

Portfolio evidence →

What we can conclude now

Evidence readiness is not equipment condition

CAPABILITY	EVIDENCE	WHAT WE CAN SAY	WHAT IS MISSING	NEXT STEP
Generator support	Limited	Generator use can be called avoidable only in blocks with complete readings	Battery output limit at 14:50	Raise action →
Service to customers	Limited	Power delivered to each feeder	What customers needed (demand)	Review service →
Battery condition	Limited	Time below the 30% reserve	Battery temperature and a capacity test	Review battery →
Fuel balance	None	No fuel balance	Delivery record and closing tank reading	Review fuel →
Backup readiness	None	Generator ran; no outage or start test seen	A witnessed start test	Review backup →

Data sources · data quality, not equipment condition

Online · Stale · Offline · Not connected

SOURCE	STATUS AT WINDOW END	QUALITY	WHAT IT PROVIDES	AFFECTS
Battery controller (PCS)	Online	Gap	One report missing at 14:50	Generator support
Generator controller	Online	Good	Status and output every 15 min	Generator support · fuel
Feeder meters (3)	Online	Good	Delivered power per feeder	Service
Solar inverter	Online	Good	Output and available power	Generator support · renewable
Fuel tank sensor	Stale	Limited	Opening reading only	Fuel balance

Most useful next evidence

1. **Restore battery output-limit reporting**
   One missing report leaves 30 minutes unclassified.
   Raise action →

2. **Measure what the clinic needs**
   Delivered power alone cannot show unmet demand.
   Review service →

3. **Test battery capacity**
   A rating plate is not a measured capacity.
   Open battery →

ⓘ Each next step can become a managed action. Record-level detail and how each limit is set are on Evidence › Analysis.

Canonical basis: v6.9 8.35 · 8.37 · **Question:** What can AssetOps substantiate, and what evidence is missing?
Preserved: Only accepted evidence is product-visible. Source quality and physical asset condition are separate.
Adapted: Window-scoped capabilities and explicit sampling denominators.
Deferred: Broad gateway management and an unexplained overall quality score.
Content owners: CapabilityReadiness · SourceHealth · AcceptedEvidence · ClaimLineage · **Prototype route:** #/evidence

Generator support merits review

F-104 · Kobo Mini-grid · generator dispatch · 6 Oct 2026

Open site

Overview | Analysis | Evidence | Alternatives | Recommendation | History

AFFECTS Supply cost per kWh · ALSO CHECK Reliable service · Battery and generator life · Seen on 24 of the last 30 days

Priority: High · Confidence: Low ⓘ · State: Open · Claim: Candidate avoidable · Lens: Cost · Kobo Mini-grid · Dispatch · 6 Oct 14:20–15:20

What happened

Selected hour · 14:20–15:20 EAT

Inspect evidence ↗

The generator remained online while PV and battery capability appeared sufficient for observed load in two supported portions.

30 minutes are classified Candidate avoidable. A further 30 minutes remain Indeterminate because a PCS capability report is missing.



Candidate runtime	Associated fuel	Cost exposure	Capability support
30 min	2.43–2.97 L	$3.31–3.97	4 / 5
Derived under declared continuity assumptions	Modelled; not fuel-meter measurement	Modelled fuel + scenario maintenance	PCS report endpoints; 2 of 4 segments supported

What limits this conclusion

Useful battery charging is an alternative explanation. A lower generator runtime can leave less stored energy for later service.

Missing capability and 25 minutes of episode transition uncertainty lower confidence. No observed override record does not prove that none existed.

See analysis and alternatives →

What to do next

Review the generator stop threshold

Test the policy with declared service, reserve and terminal-energy guardrails. Verify using a later accepted evidence window.

Review recommendation

Inspect financial assumptions →

Where to look next

1 **Analysis**
How the readings lead to the classification, block by block.
Open Analysis →

2 **Evidence**
The accepted records behind the claim, with their basis.
Open Evidence →

3 **Alternatives**
Other explanations and what would tell them apart.
Open Alternatives →

4 **Linked action**
ACT-048 · stop-setting trial, now in its verification window.
Open ACT-048 →

Canonical basis: v6.9 8.32 · 8.37 · **Question:** What is the defensible conclusion and its consequence?
Preserved: Claim, confidence and workflow state are distinct. Finding owns classification; finance only translates consequence.
Adapted: Retains the owner-approved Analysis destination and plain-language default.
Deferred: A claim of proven waste or net realized savings.
Content owners: FindingRevision · DispatchWindow · FinancialConsequence · ActionLink · **Prototype route:** #/finding

Findings › F-104 › Analysis · site: Kobo Mini-grid

Generator support merits review

F-104 · analytical method, input comparison and uncertainty

Open site

Overview | **Analysis** | Evidence | Alternatives | Recommendation | History

AFFECTS Supply cost per kWh ALSO CHECK Reliable service Battery and generator life Seen on 24 of the last 30 days

F-104 · Analysis Confidence: Low ⓘ Dispatch model dispatch-2 · Foundation v4 Inspect evidence ↗

Constraint comparison

Selected 14:20–14:35 interval

OBSERVED / DECLARED INPUT	VALUE	TEST	RESULT
Delivered load	30 kW	Compare to capability baseline	Observed
PV available	42 kW	42 ≥ 30 × 1.1	Supported
PCS discharge capability	30 kW	Both interval endpoints required	Supported
Battery SOC / reserve	74% / 30%	SOC exceeds reserve + 10 points	Supported
Minimum runtime	30 min	Already elapsed at selected interval	Not binding
Generator stop threshold	90% SOC	Reported policy explains continued operation	Alternative

Bounded fuel calculation

```
Candidate duration     0.50 h
Generator output         18 kW
Public fuel coefficient 0.30 L/kWh
Declared model band       ±10%

0.50 × 18 × 0.30 × [0.9, 1.1]
= 2.43–2.97 L
```

Fuel associated with candidate operation. It is not a counterfactual avoided-fuel calculation.

Coverage, confidence and alternatives

ISSUE	EVIDENCE INTERPRETATION	EFFECT
PCS report absent at 14:50	Two adjacent segments require the missing endpoint	30 minutes Indeterminate
Start / stop timing	15 minutes start uncertainty + 10 minutes stop uncertainty	Episode confidence lowered
Battery charging	Generator supplied energy into storage	Net savings remain unverified
Operator override	No effective override record observed in selected interval	Unobserved override not excluded

ⓘ Finding classification is produced by backend analytics. The Site card, Financials page and frontend chart must consume this result rather than re-derive it.

Canonical basis: v6.9 8.32 · 8.37 · **Question:** How do the measurements lead to the classification?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Analysis is a deliberate additional Finding contextual tab. No additional subtab row is introduced.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: DispatchIntervalRevision · ClaimLineage · AnalyticDeclaration · **Prototype route:** #/analysis

Kobo Mini-grid

Simulated site · MG-DEMO-01

Financials · technical behaviour translated to business consequence

2 consequence lines | Data confidence: Low ⓘ | USD · selected hour · no financial status here; status stays on the source findings | Portfolio roll-up →

Operating-cost exposure	Revenue / service	Asset exposure	Realized savings
$3.31–3.97	**Unavailable**	**Scenario only**	**Not established**
Modelled fuel + scenario maintenance accrual	No independently supported unserved-energy basis	Battery replacement covered by warranty to 2034 if terms are kept	Counterfactual energy and service need verification
		View detail →	

What happened → what it means in money

Translated once here and reused by Overview, Finding and Portfolio · select a source to return to its finding

WHAT HAPPENED	HOW IT BECOMES MONEY	CONSEQUENCE	BASIS	CONFIDENCE	SOURCE
2.43–2.97 L fuel during avoidable generator time	Fuel at $1.20/L	$2.91–3.57	Modelled	Low ⓘ	F-104 ↗
0.50 h avoidable generator time	Maintenance allowance at $0.80/h	$0.40	Scenario	Low ⓘ	F-104 ↗
Generator output charged the battery	Stored energy may replace later supply	Net savings unverified	Unknown	—	View constraint ↗

Assumptions in use

Fuel price	**$1.20 / L**
Maintenance accrual	**$0.80 / runtime hour**
Currency	**USD**
Effective period	**Selected assessment window**
Interpretation	**Illustrative scenario inputs**

Edit example assumptions

Trade-offs and exclusions

Stored energy matters. Generator output charged the battery. Lower runtime may shift energy supply into a later period.

No monthly extrapolation is made from this hour. Fuel residuals, battery exposure and alternative scenario worlds are not added into this total.

Inspect technical basis ↗

ⓘ Financials carry no status of their own. Exposure is not a statement of net or realised savings, and lines with different bases are not added together.

Canonical basis: v6.9 8.36 · 6 · **Question:** What is the bounded economic consequence?
Preserved: Site owns the technical bridge. Global pages aggregate its results; financial pages have no assessment enum.
Adapted: Two real consequence lines and editable assumptions instead of a broad finance suite. Subtabs Overview | Assumptions; v6.9 Drivers is the bridge on Overview; Scenarios is a placeholder with an empty state (owner, follow-up 3). Implementation note: ranges use outward rounding; totals use unrounded amounts.
Deferred: Cash flow, currency conversion and monthly extrapolation.
Content owners: FinancialConsequenceRevision · FinancialAssumptionSet · **Prototype route:** #/financials

Sites › Kobo Mini-grid › Financials

Kobo Mini-grid

Simulated site · MG-DEMO-01

Financials · assumptions and interpretation history

Interactive mockup: changing the fuel price updates example consequence values across these screens. It does not write to the product database.

Financial assumptions

Site-scoped business context · BC-MG-01

Fuel price · USD per litre

1.20

Example: change 1.20 to 1.50

Maintenance accrual · USD per hour

0.80

Scenario accrual, not a maintenance invoice

Currency

USD

Effective from

6 Oct 2026 · 14:20 EAT

A production save appends a version, preserving the previous assumption / consequence pair. Technical evidence and classification remain unchanged.

Apply example price Return to financial overview

Impact preview

CANDIDATE-WINDOW COST EXPOSURE

$3.31–3.97

Fuel range: **2.43–2.97 L**

Runtime: **0.50 h**

Classification: **Candidate avoidable**

Confidence: Low

Technical result unchanged

Interpretation history

VERSION	FUEL PRICE	MAINTENANCE	PROVENANCE	STATUS
BC-MG-01-v1	$1.20/L	$0.80/h	Illustrative initial assumptions	Retained
Proposed next version	$1.20/L	$0.80/h	Reviewer example edit	Mockup only

Canonical basis: v6.9 8.36 · **Question:** Which assumptions turn this quantity into money?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: One interactive price edit demonstrates the technical/economic boundary.
Deferred: Real database writes and shared portfolio assumption sets.
Content owners: FinancialAssumptionSet · FinancialConsequenceRevision · **Prototype route:** #/assumptions

Generator support merits review

F-104 · proposed policy intervention and declared success criterion

Open site

AFFECTS Supply cost per kWh ALSO CHECK Reliable service Battery and generator life Seen on 24 of the last 30 days

F-104 · Recommendation Policy review; not a diagnosis of controller malfunction

Proposed intervention

Trial a lower generator stop threshold.

Review moving the stop threshold from **90% to 70% SOC** while retaining the 30% reserve and 30-minute minimum runtime. Record the applied configuration and observe a later comparable period.

Suggested owner	**Site controls lead**
Action type	**Policy review / supervised trial**
Origin	**F-104 · retained analytic revision**
Configuration change	**CHG-019**

Accept example recommendation

Declare success before evaluating

ROLE	CRITERION
Target	Candidate runtime reduced by at least 25%
Guardrail	No new battery reserve breaches
Guardrail	No increase in critical service deficit
Guardrail	Comparable terminal storage and energy context
Evidence	Adequate matched-window coverage; confounders reviewed

ⓘ Acceptance creates managed work. It does not close the Finding, verify the change or send a real controller command.

Evidence needed to verify

1 **Applied configuration record**
Time-valid policy change and authority retained.
Inspect change →

2 **Later operating evidence**
Accepted reports in a later operational window; comparison worlds stay separate.
Inspect verification →

Canonical basis: v6.9 8.32 · 8.38 · **Question:** What action should be accepted, and how will success be judged?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: Recommendation · Action · VerificationPlanVersion · **Prototype route:** #/recommendation

Actions › ACT-048 › Verification · site: Kobo Mini-grid

Reduce extended generator support

ACT-048 · source F-104 · owner: site controls lead

Open site

Overview | Plan | Execution | Evidence | **Verification** | History

ACT-048 · Verification window · Provisional target pass · Final outcome pending adequate closed-window evidence

OWNER	DUE / SLA
Site controls lead	Window close · not set
STATE	**SOURCE**
Verification window	F-104
SUCCESS CRITERION	
≥25% less avoidable generator time; no reserve or service trade-off	

Target change
−100%
30 → 0 min candidate runtime · provisional

Reserve breaches
0 → 0
Covered portions; guardrail passes so far

Critical service
Incomplete
Requested-service evidence still missing

Final outcome
Pending
A passed target alone cannot verify success

Target and guardrails evaluated together

Rule fixed before evaluation · short-window simulated example

METRIC	ROLE	BEFORE	AFTER	EVIDENCE	RESULT
Candidate runtime	Target	30 min	0 min	Later-window dispatch revision	Pass so far
Reserve-breach duration	Guardrail	0 min	0 min	SOC + effective reserve policy	Pass so far
Critical unserved energy	Guardrail	Unknown	Unknown	Load request baseline absent	Not assessable
Terminal stored energy	Guardrail	Not comparable	Not comparable	Equivalent horizon not yet established	Needs evidence

Activity evidence

Policy application record	Captured
Execution state	**Completed**
Change reference	**CHG-019**
Owner	**Site controls lead**

Completed work is evidence of activity, not resolution.

Outcome evidence

If this window closes without service and stored-energy evidence, the outcome is **Inconclusive**. The Finding remains open.

A later adequately covered, comparable attempt may be Verified. Each attempt retains its own criteria and evidence.

Review a complete evidence example

Canonical basis: v6.9 8.38 · 8.11 · **Question:** Did the action work without harmful trade-offs?
Preserved: Target plus guardrails and sufficient closed-window evidence determine outcome; activity completion cannot resolve a Finding.
Adapted: Shows the honest incomplete state by default, with a separate completed-evidence review example.
Deferred: Real equipment actuation and field-validated efficacy.
Content owners: Action · VerificationPlanVersion · VerificationOutcome · **Prototype route:** #/verification

Kobo battery

BAT-001 · 200 kWh nominal · site: Kobo Mini-grid

Open site

Watch · Capacity: unverified · Battery condition evidence · separate 30-day illustration · Inspect condition evidence ↗

Reserve exposure	Thermal exposure	Power stress	Usable capacity
5 breaches	Unknown	3 events	Unverified
Observed in covered periods	No accepted temperature series	Above declared high-power threshold	200 kWh is a configuration rating

Observed reserve and power exposure

Illustrative event count · accepted SOC and effective reserve policy

Week 1	1 event
Week 2	2 events
Week 3	1 event
Week 4	1 event

A reserve breach is a policy exposure event. It is not a direct measurement of capacity loss.

Interpretation and next evidence

Observable reserve exposure merits review. The history is insufficient to infer lifetime accumulated stress or a precise remaining-life estimate.

1 **Verify usable capacity**
A controlled test can replace the current unverified capacity assumption.
Create test action →

2 **Add thermal evidence**
Temperature exposure is unknown, not normal.
Review capability gap →

Condition evidence and claim boundary

DIMENSION	EVIDENCE BASIS	WHAT MAY BE CONCLUDED	WHAT REMAINS UNKNOWN
Reserve discipline	SOC + time-valid policy	Frequency and duration of observed excursions	Unobserved periods
High-power exposure	Reported battery power + capacity assumption	Modelled stress exposure	Validated degradation rate
Capacity	Nameplate / configured rating only	200 kWh declared nominal rating	Measured usable capacity
Asset-life consequence	Scenario assumptions only	Potential replacement exposure	Remaining years or replacement date

ⓘ Warranty: cover to 1 Mar 2034 · at risk because temperature cannot be shown · Warranty →. No precise remaining-life prediction is supported.

Canonical basis: v6.9 8.27 · **Question:** What condition and asset-life risk can we substantiate?
Preserved: Asset Health owns condition; Site Health only summarizes. Unknown historical stress stays unknown.
Adapted: Lifecycle risk is a bounded section in Health rather than a new root.
Deferred: Precise RUL, Warranty and validated degradation prediction.
Content owners: AssetHealthSummary · BatteryStressWindow · CapacityEvidence · **Prototype route:** #/asset

Kobo Mini-grid

Simulated site · MG-DEMO-01

Health · site-level asset-condition summary

Asset assessment: Watch — 3 monitored assets · summary only; asset detail lives in Assets

Physical asset condition

Opening an asset moves you to Assets; the breadcrumb keeps this site

ASSET	ASSESSMENT	EVIDENCE	MAIN REASON	OPEN FINDINGS	OPEN WORK	
Battery · BAT-001	Watch	Limited	Time below reserve; capacity not measured	0	0 · capacity test recommended	Open asset →
Generator · GEN-001	Unknown	Running time only	Condition cannot be judged from running time	1 · F-104	1 · ACT-048	Open asset →
Solar inverter · INV-001	Unknown	Output and available power	No condition assessment	0	0	Open asset →

Data sources · data quality, not equipment condition

Online · Stale · Offline · Not connected

SOURCE	STATUS AT 15:20	QUALITY	AFFECTS	OPEN
Battery controller (PCS)	Online	Gap	Generator support classification	Site Evidence →
Battery temperature	Not connected	None	Battery condition	Asset Health →
Site gateway	Online	Good	Freshness of all sources	Site Evidence →

ⓘ A gateway being Online would not make a battery Healthy. Physical condition and data quality use different owners and vocabularies.

Canonical basis: v6.9 8.19 · **Question:** Which physical assets need attention?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: AssetHealthSummary · SourceHealth · **Prototype route:** #/health

Kobo · Gateway & Ingestion

Simulator Lab · source-side proof workspace · illustrated run R-MG-01

Gateway & Ingestion

R-MG-01 · Committed | Ingestion completed | Frozen inputs · mapping v1 · envelope schema 2 | Open in AssetOps →

1 · Private world
SOC at T: 74%
Accepted charging: 30 kW

→

2 · Device report
PCS SOC: 74%
Source time: 14:20:00

→

3 · SourceEnvelope
Mapping v1 · schema 2
Published: 14:20:03

→

4 · Accepted evidence
Validation + receipt
Received: 16:00:00

Selected canonical envelope

Exactly the source boundary; no true-SOC or expected-Finding field

```
device_id: battery-pcs-controller
signal_id: state-of-charge
site_id: MG-DEMO-01
source_time: 2026-10-06T11:20:00Z
published_at: 2026-10-06T11:20:03Z
family: MEASUREMENT
measurement: STATE_AT_INSTANT
value: "74"
unit: "%"
quality: GOOD
mapping: simulator-lab-gateway-dispatch / v1
```

Compact field projection. The canonical wire document is nested and carries a computed message identity. received_at is assigned by ingestion, never the gateway.

Truth versus observed

AT / SIGNAL	PRIVATE TRUTH	OBSERVED	PRODUCT CONSEQUENCE
14:20 · SOC	74%	74%	Accepted SOC
14:35 · SOC	77.5625%	Published 2 min late	Latency retained
14:50 · PCS limit	30 kW	No report	Two segments indeterminate

Release and ingestion

Commit	**C-MG-01 · immutable manifest**
Batch	**B-MG-01**
Stage watermark	**Complete through selected release**
Duplicate retry	**No duplicate accepted fact**
Conflicting bytes	**Original retained; conflict recorded**

Inspect accepted record

What never crosses

Private scenario cause, true state, controller intent, accepted flow set, noise draw and expected Finding remain in this workspace.

The gateway observes, maps and publishes. Normal ingestion and analytics determine acceptance and conclusions.

Follow evidence to Dispatch →

SEPARATE WORKSPACE

→ Open in AssetOps

Demo workspace
3 sites · illustrative records

Canonical basis: v6.9 8.12–8.17 · especially 8.13 / 8.17 · **Question:** How did physical state become accepted product evidence?
Preserved: Private truth stays gated. Commit seals source envelopes; ingestion alone creates accepted evidence.
Adapted: The existing cockpit becomes an explicit four-stage proof view.
Deferred: Broad authoring, outage catalogue and real protocol adapters.
Content owners: LabProjection · DeviceObservation · SourceStage · CommitManifest · BatchProgress · **Prototype route:** #/lab

GreenHarvest Cold Room

Simulated site · CR-DEMO-01

Produce cold room · 2–4°C policy · hybrid: 15 kWp solar, 30 kWh battery, grid & diesel backup

Degraded · Confidence: High · Potential product exposure · Power → backup → cooling chain fully reconstructed · no spoilage determination · Inspect evidence ↗

Objective: keep the chamber at 2–4°C through power, cooling and operational disturbances.

Stock kept saleable

Needs attention

25–35 min above 4°C · peak 8.2°C

Exposure found · stock impact unknown

View detail →

Cooling continuity

Needs attention

Automatic start failed · 3 repair visits without a result

Repair done · start not proven

View detail →

Compressor life

Watch

Recovered after power returned · start test overdue, coil cleaning due

Early warning · condition not checked

View detail →

Evidence quality

Watch

Loading and door records not connected

Gap known

View detail →

Not measured yet

Paid cold-room use, Cost per cooling unit, Market value of produce

Needs occupancy, cooling energy and sales records

What happened · power → cooling → product conditions

Incident INC-203 · chronology from accepted readings

Open incident →

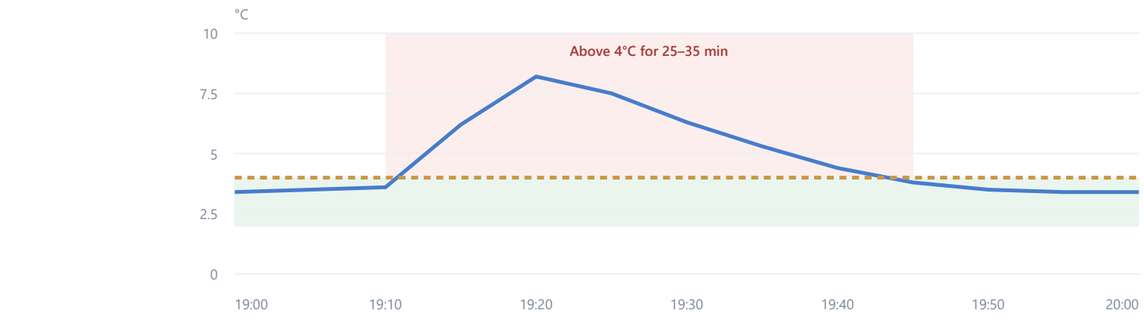
19:11	19:11	19:11:05	19:11:10	19:14	19:15	19:19	by 19:45
Grid supply lost	**Compressor stopped; battery holds controls**	**Automatic start requested**	**Generator failed to start**	**Started by hand**	**Cooling running again**	**Peak 8.2°C**	**Back within 2–4°C**
Site power meter	By design at night · inverter-charger	Generator controller	Generator controller	Operator log	Compressor meter	Chamber probe	Chamber probe

Grid power was lost after dark, when solar was not producing. As designed, the battery kept controls and alarms running while the compressor waited for the generator — but the generator did not start on its own. Cooling stopped for four minutes until it was started by hand, and the chamber stayed above 4°C for 25–35 minutes.

Chamber temperature

Operating conditions, not an inventory-loss model

Single line diagram →



Policy band 2–4°C. Shaded area: time above the limit, bounded by 5-minute readings.

Needs attention

1 **ACT-051 · Witness an automatic-start test**
The starter inspection is done, but no test shows backup works again.
Actions › ACT-051 →

2 **WR-019 · Inspection done, outcome unverified**
Completed work is activity, not a fix.
Work › Maintenance →

3 **F-118 · Keep the finding open**
Stays open until the declared start test passes.
Findings › F-118 →

Bounded money consequence

Not priced

Potential product exposure. Stock value and product response are not declared, so **no loss figure is stated**.

Open financial bridge →

Next step

Witness an automatic-start test

Cut grid supply under supervision and record start request, generator start, power to the compressor and chamber temperature.

Review action ACT-051

Canonical basis: v6.9 8.28 · **Question:** Were product conditions protected through the disturbance?
Preserved: Temperature, backup, cooling, maintenance and evidence remain distinct; exposure is not spoilage.
Adapted: Same overview skeleton as the mini-grid: status with confidence → objective → canonical cards → causal strip → context + needs attention (with destinations) → bounded consequence + next step. Evidence card uses evidence vocabulary (Limited) in neutral grey rather than v6.9 Healthy.

Deferred: Spoilage diagnosis and stock ERP.

Content owners: TemperatureExcursion · BackupEventChain · Incident · WorkSummary · **Prototype route:** #/cold

GreenHarvest Cold Room

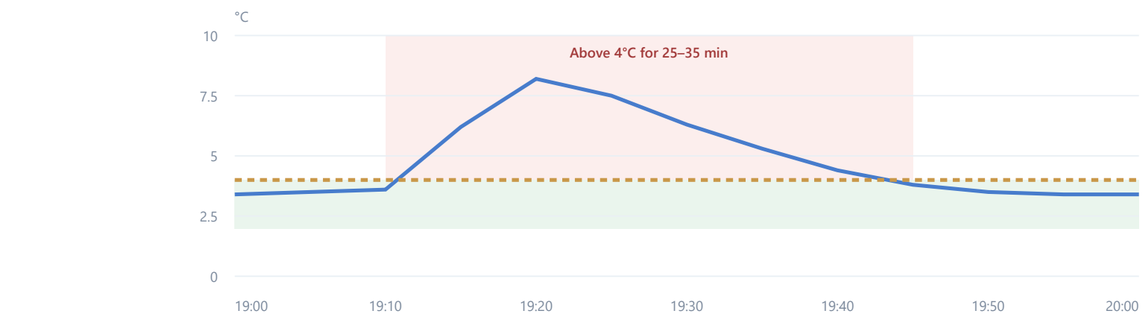
Simulated site · CR-DEMO-01

Performance · temperature policy and bounded exposure

Needs attention · Coverage: 13 / 13 readings · Policy 2–4°C · one chamber probe · readings every 5 min · in policy 42–58% of the hour · Inspect temperature claim ↗

Time above 4°C	Peak	Exposure size	Product outcome
25–35 min	8.2 °C	50–90 °C·min	Unknown
Between the last in-policy and first out-of-policy readings	+4.2°C above the 4°C limit	Area above 4°C, bounded by the 5-minute readings	Temperature exposure is not spoilage

Temperature with operating context



19:11 **Power lost** — Cooling interrupted

19:15 **Cooling restored** — Compressor meter

19:19 **Peak observed** — 8.2°C

Excursion context

Excursion window	Began 19:10–19:15 · ended 19:40–19:45
Upper policy limit	4°C
Peak observed	8.2°C at 19:19
Power outage	Observed 19:11 · backup failed →
Door opening	No record · door log not connected
Loading / batch	Not established
Defrost	No record observed

Open Incident INC-203 →

ⓘ Durations are bounded by the 5-minute reading spacing. Exposure size is bounded the same way (lowest and highest reading in each 5-minute step); it does not describe what happened to the product.

Claim boundary

The accepted probe and power records support a temperature excursion during a backup interruption. Product-specific quality or spoilage conclusions require product identity, exposure policy and additional evidence.

Canonical basis: v6.9 8.29 · **Question:** How far and how long did temperature exceed policy?
Preserved: Excursions and exposure remain bounded by actual temperature evidence and configured policy.
Adapted: Duration, in-policy share and exposure size are computed from the 5-minute readings as bounded ranges (duration bracketed by readings; °C·min bounded by the lower and upper reading in each 5-minute step (50–90), replacing the earlier ≈70 straight-line point estimate) rather than v6.9 point values.
Deferred: Spoilage determination and unverified exposure valuation.
Content owners: TemperatureExcursion · PolicyVersion · AcceptedEvidence · **Prototype route:** #/temperature

GreenHarvest Cold Room

Simulated site · CR-DEMO-01

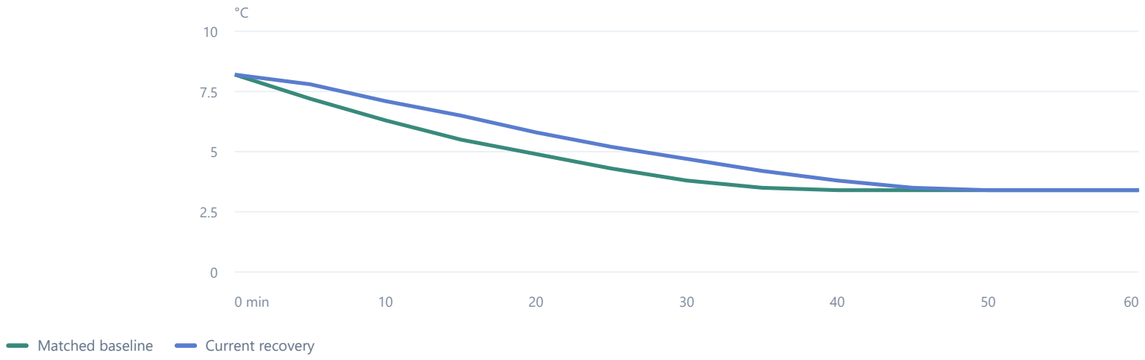
Performance · recovery under comparable conditions

Comparison limited · Confidence: Low ⓘ · Illustrative recovery curves · loading context not matched · Inspect recovery basis ↗

Usual time back to 4°C	This time back to 4°C	Difference	Comparable?
36 min	42 min	+17%	Not yet
Earlier comparable recoveries · illustrative	Recovery after cooling restarted · illustrative	A description, not a diagnosis	Loading and door records missing

Recovery curves

Example curve shapes; selected event-to-policy recovery



Context normalization

FACTOR	EVIDENCE	MATCH
Ambient temperature	29–32°C	Matched
Power availability	Continuous during recovery	Matched
Door-open duration	Limited record	Limited
Loading / batch	Not established	Unknown
Defrost	No event observed	Not excluded

ⓘ These examples do not support a definitive cooling-degradation or compressor-failure conclusion. Obtain comparable loading and door context before upgrading the claim.

Next best evidence

1. **Repeat a comparable recovery test**
   Record load, door opening, ambient conditions, defrost and cooling energy.
   Plan a witnessed test →

2. **Inspect the excursion chain**
   Check whether lost power, rather than refrigeration condition, explains the event.
   Open Backup →

Canonical basis: v6.9 8.30 · **Question:** Is recovery slower under comparable operating conditions?
Preserved: Context normalization precedes a degradation claim.
Adapted: Shows an evidence-limited comparison so similar curves cannot imply a false diagnosis.
Deferred: Automatic compressor diagnosis.
Content owners: CoolingRecoveryWindow · ContextMatch · ClaimLineage · **Prototype route:** #/cooling

Sites › GreenHarvest Cold Room › Performance

GreenHarvest Cold Room

Simulated site · CR-DEMO-01

Performance · grid outage to cooling-service recovery

Overview | Foundation | Health | **Performance** | Findings | Work | Financials | Evidence

Temperature | Cooling | **Backup**

Degraded · Confidence: High · One outage in the selected hour · automatic start failed · recovered by hand · Open INC-203 →

What the chain shows

The backup failed; the refrigeration did not.

The battery held controls and alarms as designed. The compressor restarted within a minute of generator power reaching it at 19:15, and the chamber was back within policy by 19:45. The evidence points to the generator's automatic start, not the cooling plant or the solar-battery system. Which starter part failed still needs inspection.

Grid-loss event	Automatic start	Manual recovery	Cooling resumed
19:11	Failed	19:14	19:15
Site power evidence	Command and controller status observed	Operational record	Compressor meter / status

Reconstructed event chain

19:11	19:11	19:11:05	19:11:10	19:14	19:15	19:19	by 19:45
Grid supply lost	**Compressor stopped; battery holds controls**	**Automatic start requested**	**Generator failed to start**	**Started by hand**	**Cooling running again**	**Peak 8.2°C**	**Back within 2–4°C**
Site power meter	By design at night · inverter-charger	Generator controller	Generator controller	Operator log	Compressor meter	Chamber probe	Chamber probe

Backup event chains

Same chain as the mini-grid Backup view · no other outage in the selected hour

WHEN	GRID	START REQUESTED	GENERATOR	POWER TO LOAD	LOAD RESPONSE	OUTCOME FOR SERVICE	RESULT
6 Oct 19:11	Lost	Seen 19:11:05	Start failed	Generator path restored 19:14 by hand · transfer record missing	Compressor shed 19:11 (battery holds controls) → running 19:15	Above 4°C for 25–35 min	Fail

Evidence by causal step

STEP	ACCEPTED OBSERVATION	WHAT IT ESTABLISHES	LIMIT
Grid supply	Lost at 19:11	Supply interruption	Not the root cause of controller failure
Start command	Seen at 19:11:05	An automatic start was requested	Does not identify the defective part
Generator	Failure status at 19:11:10	Start did not succeed	Starter fault requires investigation
Power / compressor	Cooling resumes at 19:15	Service was restored	Transfer-path detail may remain incomplete
Temperature	Peak 8.2°C at 19:19	Bounded exposure consequence	No spoilage determination

ⓘ The backup chain and temperature consequence share evidence references. A maintenance completion leaves the Finding open until a declared outcome test passes.

Canonical basis: v6.9 8.31 · **Question:** What failed between loss of grid and restored cooling?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: States the 8.31 discrimination (backup versus refrigeration failure) in one plain sentence; shares the backup chain table with the mini-grid; the missing transfer record is shown as an explicit missing link.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: BackupEventChain · Incident · TemperatureExcursion · **Prototype route:** #/coldbackup

Depot Charging Hub

Simulated site · EV-DEMO-01

2 × 60 kW connectors · 80 kW site limit · no stationary battery

Needs attention · Confidence: Medium ⓘ · Assessed: session service, availability and site power · cause of the interruption not yet established · Inspect evidence ↗

Objective: let depot vehicles charge when scheduled, within the 80 kW site import limit.

Buses ready to run

Needs attention

1 interrupted · 1 unknown of 12 sessions · connector 2 out 1 h

Connector check open

View detail →

Room before expansion

On track

Peak 76 kW of the 80 kW limit

Within the limit · little room left

View detail →

Charger use

Watch

120 kWh in an 8-hour day · under a fifth of capacity

Opportunity seen

View detail →

Charger life

Watch

Fault on connector 2 · inspection overdue since 30 Sep

Early warning · condition not checked

View detail →

Evidence quality

Watch

One session end and 1 h of status missing

Gap known

View detail →

Not measured yet

Energy cost per km, Energy billed against metered

Needs odometer readings and settlement records

What happened on the service day

Plain sequence from accepted charger and meter records

Open Sessions →

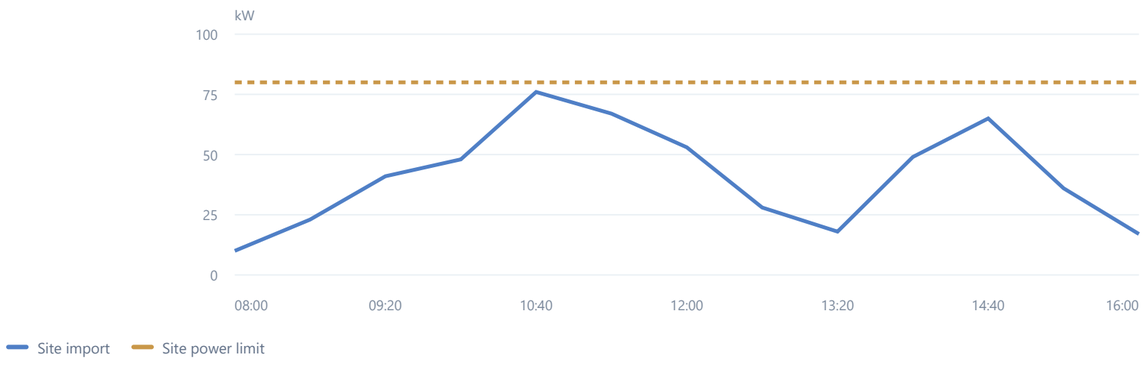
08:00	10:40	14:03	14:11	14:11–15:00	15:00–16:00	15:04
Charging day starts	**Site peak 76 kW**	**Session SES-011 starts**	**A fault stops the session**	**Connector 2 out of service**	**Connector 2 status missing**	**Session SES-012 starts; no end record**
Both connectors available	Under the 80 kW limit	Connector 2	Connector 2 · 3.6 kWh delivered	Status reports	No status reports	Connector 2

Connector 2 stopped a session on a fault and was then out of service for about an hour. Site power stayed under its limit, so the site supply did not cause it; the charger or vehicle cause is not yet diagnosed.

Charging activity and site power

08:00–16:00 · two connectors · context, not a status source

Single line diagram →



Needs attention

1. **F-130 · Connector 2 stopped a session**
   Fault event and meter record agree; 3.6 kWh delivered.
   Findings › F-130 →
2. **One session outcome unknown**
   No end record for SES-012; it counts as neither success nor failure.
   Performance › Sessions →
3. **Connector 2 status missing for 1 h**
   Availability stays a range until the history is recovered.
   Evidence →

Bounded money consequence

$36.00

Gross tariff estimate for 120 kWh delivered in complete sessions at a $0.30/kWh scenario tariff. **Not cash collected.** The interrupted session is not priced: unmet demand is not established.

Open financial bridge →

Next step

Check connector 2 and witness a full session

Retrieve the fault log for SES-011, then run a complete session on connector 2 with start, end and meter records.

Review recommendation

Canonical basis: v6.9 2.2 / 2.4 pattern · EV extrapolation · **Question:** Can drivers charge, and what constrained their service?
Preserved: Stable Site / Asset / Finding shell, evidence ceilings and bounded financial translation.

Adapted: EV is extrapolated (v6.9 has no EV section). Uses the shared site-overview skeleton; sessions, connector availability and site power replace mini-grid dispatch content; cards use canonical assessments only.
Deferred: Roaming, billing settlement and vehicle-battery health.
Content owners: SessionServiceWindow · ConnectorAvailabilityWindow · ChargingEnergyWindow · **Prototype route:** #/ev

Depot Charging Hub

Simulated site · EV-DEMO-01

Performance · charging sessions and delivered service

Service needs attention — 12 attempts · keep incomplete records separate from observed outcomes — Inspect session lineage ↗

Attempts	Confirmed delivery	Fault-aborted	Incomplete
12	10	1	1
Accepted session starts	Completed with delivered-energy evidence	Observed fault and stop event	End record absent; outcome unknown

Session service records

120 kWh is the summed metered energy in complete records, including the aborted session; the incomplete session is excluded

SESSION / CONNECTOR	PERIOD	ENERGY	OUTCOME	EVIDENCE COMPLETENESS	NEXT STEP
SES-010 · connector 1	13:12–13:46	18.4 kWh	Delivered	Start, end and meter records	Inspect records ↗
SES-011 · connector 2	14:03–14:11	3.6 kWh	Fault-aborted	Fault, stop and meter records	**Review connector →**
SES-012 · connector 2	15:04–unknown	Unknown	Incomplete	End record absent	Inspect missingness ↗
9 earlier completed sessions	08:00–13:05	98.0 kWh	Delivered	Completed records present	**Energy summary →**

How to read the denominator

10 confirmed completions + 1 observed failure + 1 unknown = 12 attempts. Do not collapse unknown into a failure or present a success percentage without stating its denominator.

Cause remains bounded

A fault event establishes interruption. Low charging power alone would not establish site-power constraint: vehicle acceptance, requested current and control-limit records are needed.

Canonical basis: v6.9 2.2 pattern · 8.32 / 8.35 · EV extrapolation · **Question:** Which sessions completed, failed or remain unknown?
Preserved: Outcomes are evidence-derived, with unknowns preserved.
Adapted: New EV read models use canonical Finding and Evidence detail.
Deferred: Billing settlement and unobserved session energy.
Content owners: ChargingSession · SessionServiceWindow · AcceptedEvidence · **Prototype route:** #/sessions

Depot Charging Hub

Simulated site · EV-DEMO-01

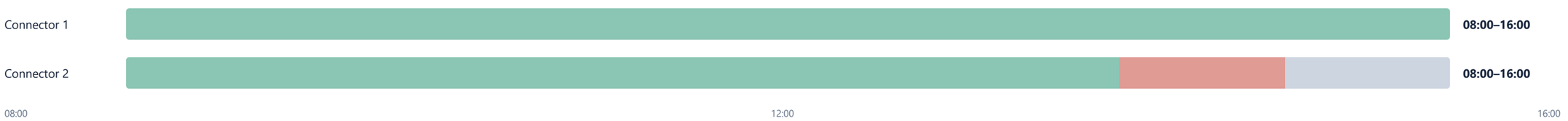
Performance · connector availability and unknown time

Evidence limited — 2 connectors × 8 scheduled hours = 16 connector-hours

Known available	Known unavailable	Unknown	Known-time availability
14 connector-h	1 connector-h	1 connector-h	93.3%
State-supported service time	Faulted or unavailable state evidenced	No trustworthy state coverage	14 / 15 known hours; coverage 93.75%

Connector states over the service schedule

Available · Unavailable · Unknown



Connector 2: 6 h available, 1 h observed unavailable and 1 h unknown. Aggregated spans are illustrative; production uses accepted transition timestamps.

Full-schedule availability bound

87.5–93.75%

Lower bound: 14 / 16 hours. Upper bound: (14 + 1 unknown) / 16. Unknown time is not silently included as available.

Most useful next evidence

Recover the connector status/heartbeat history for the missing hour. A telemetry outage does not prove the charging connector was physically unavailable.

Inspect connector-state lineage ↗

Canonical basis: v6.9 8.35 pattern · EV extrapolation · **Question:** Was each connector available during its intended service hours?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: A scheduled-service denominator replaces mini-grid served-load semantics.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: ConnectorAvailabilityWindow · ServiceSchedule · **Prototype route:** #/availability

Depot Charging Hub

Simulated site · EV-DEMO-01

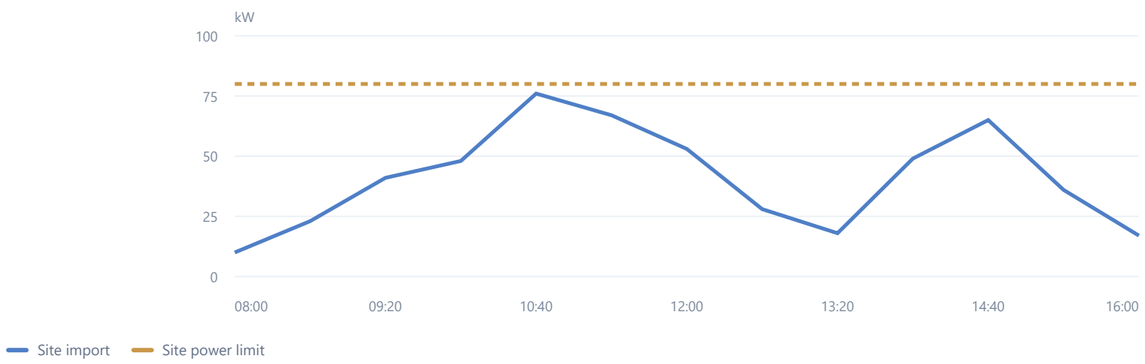
Performance · charging energy and site-power context

Energy basis: measured — Complete session meter records · incomplete session energy excluded — Inspect meter basis ↗

Metered session energy	Observed peak import	Declared site limit	Gross tariff estimate
120 kWh	76 kW	80 kW	$36.00
Includes 3.6 kWh from aborted session	Selected 8-hour window	Time-valid configuration	Scenario tariff: $0.30/kWh

Import power and configured limit

Observed import is not proof of vehicle-requested power



What can be concluded

Observed peak below limit	Yes
Site limit caused interruption	Not established
Requested vehicle power	Incomplete evidence
Stationary battery	Not configured
Billing settlement	Not connected

Revenue is an estimate from metered kWh and the declared tariff, not payment evidence.

Technical-to-financial bridge

QUANTITY	ASSUMPTION	RESULT	LIMIT
120 kWh complete-record delivery	$0.30/kWh scenario tariff	$36 gross estimate	No collections or profit claim
Incomplete session	No complete meter interval	Unavailable	No extrapolated energy silently added
Interrupted service	No demand/sales baseline	Unpriced exposure	No invented lost revenue

Canonical basis: v6.9 8.36 pattern · EV extrapolation · **Question:** How much charging energy was delivered, and what power limits mattered?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: ChargingEnergyWindow · SitePowerLimit · FinancialConsequence · **Prototype route:** #/energy

Incidents › INC-203 › Timeline · site: GreenHarvest Cold Room

Backup failure and temperature excursion

INC-203 · GreenHarvest · power-to-product-condition case

Open site

Timeline

INC-203 · Awaiting evidence | Severity: High | Work completed; outcome unverified | Site: GreenHarvest Cold Room

Return to site →

OWNER	DUE / SLA
Maintenance lead	Escalated: site and operations
STATE	**SOURCE**
Awaiting evidence	Generator start failure · 6 Oct 19:11
CLOSURE CRITERION	
Witnessed automatic start restores cooling within policy; no repeat excursion	

Reconstructed timeline

Accepted events establish chronology; causal diagnosis remains bounded

19:11	19:11	19:11:05	19:11:10	19:14	19:15	19:19	by 19:45
Grid supply lost	**Compressor stopped; battery holds controls**	**Automatic start requested**	**Generator failed to start**	**Started by hand**	**Cooling running again**	**Peak 8.2°C**	**Back within 2–4°C**
Site power meter	By design at night · inverter-charger	Generator controller	Generator controller	Operator log	Compressor meter	Chamber probe	Chamber probe

Linked conclusions

FINDING	CLAIM	EVIDENCE
F-118 · Backup response failure	Automatic start failed after accepted command	Inspect event chain ↗
F-119 · Temperature excursion	Peak 8.2°C against 4°C upper policy	Temperature →

Work and next evidence

1 **WR-019 · Starter/control inspection**
Activity completed. Witnessed auto-start result not recorded.
Maintenance verification →

2 **ACT-051 · Witnessed backup test**
Success requires command, start, power and service recovery evidence.
Actions queue →

ⓘ Product exposure is consequence context. The incident does not assert spoilage, a diagnosed failed component or verified resolution.

Canonical basis: v6.9 8.33 · 8.31 · **Question:** What happened, what is known, and what work remains?
Preserved: Incident root retains site context; work and Finding records are linked, not copied.
Adapted: One narrow event-chain case is sufficient for the demo.
Deferred: General incident correlation engine.
Content owners: Incident · BackupEventChain · FindingLink · MaintenanceWorkRequest · **Prototype route:** #/incident

Maintenance › WR-019 › Verification · site: GreenHarvest Cold Room

Starter and control-circuit inspection

WR-019 · GreenHarvest · owner: maintenance lead

Open site

Overview | Maintenance record | **Verification** | History

WR-019 · Completed | Outcome: unverified | Linked case INC-203 · Finding F-118 remains open

OWNER Maintenance lead	**DUE / SLA** Witness test · not set
STATE Completed · unverified	**SOURCE** INC-203 · F-118
SUCCESS CRITERION Automatic start restores cooling within policy on a witnessed outage; no repeat failure	

Work evidence · activity

Completion proves the inspection happened

EVIDENCE	STATUS	SOURCE
Inspection record	Captured	Technician work record
Starter connection check	Captured	Work attachment
Configuration readback	Partial	Controller record
Witnessed auto-start test	Missing	Test not yet recorded

Post-work evidence · outcome

Resolution requires a declared test and covered follow-up window

METRIC	REQUIRED CRITERION	CURRENT RESULT
Backup start	Command followed by successful start	Not tested
Power path	Supply restored to cooling system	Not tested
Temperature	Recovery under declared policy	Not established
Recurrence	No qualifying failure in required window	Window pending

Next action

Run a witnessed automatic-start test.

Capture the start command, generator state, restored power and cooling response. Evaluate the target and guardrails under the frozen verification plan.

Open originating case | Inspect required outcome evidence ↗

ⓘ Completed work ≠ verified resolution. Missing evidence at a closed evaluation window yields Inconclusive; it cannot be auto-closed as successful.

Canonical basis: v6.9 8.34 · **Question:** Did the completed work restore the intended outcome?
Preserved: Actions and Maintenance share verification grammar but retain separate root objects.
Adapted: Two side-by-side panels make activity versus outcome unmistakable.
Deferred: Procurement, spare parts and full CMMS scheduling.
Content owners: MaintenanceWorkRequest · VerificationPlan · VerificationOutcome · **Prototype route:** #/workverification

Kobo Mini-grid

Simulated site · MG-DEMO-01

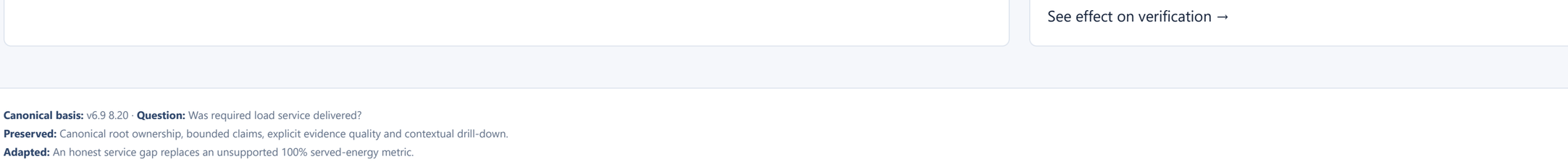
Performance · delivered load and service claim limits

Service assessment: Unknown Delivered power is observed; requested-service baseline is incomplete

Delivered load	Interruptions seen	Clinic (critical)	Unserved energy
30 kW	None	4 kW	Unknown
Sum of three feeder meters	In readings only; 40 of 60 min between readings	Delivered; what it needed is not measured	Needs demand or request evidence

Where the power went

Delivered is measured · needed is not



Delivered power per feeder, average of the selected hour. The hatched bar is the missing half of a service judgement.

Delivered power by addressed load

LOAD	ROLE	OBSERVED POWER	REQUIRED-SERVICE EVIDENCE	ASSESSMENT
Residential feeder	General service	18 kW	No supported demand baseline	Unknown
Clinic	Critical service	4 kW	Instrumented request source not yet available	Unknown
Grain mill	Productive use	8 kW	Schedule/request evidence required	Unknown

What is supported

The three meters report delivered power. These values describe supply to the loads during measured intervals; they do not independently establish what customers required.

What unlocks a service conclusion

For controllable clinic loads, pair a reported requested-power record with delivered power. For passive loads, use evidenced service availability or a labelled demand model.

See effect on verification →

Canonical basis: v6.9 8.20 · **Question:** Was required load service delivered?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: An honest service gap replaces an unsupported 100% served-energy metric.
Deferred: Precise unserved kWh without requested-service evidence.
Content owners: ServiceWindow · LoadRequestEvidence · AcceptedEvidence · **Prototype route:** #/service

Sites › Kobo Mini-grid › Performance

Kobo Mini-grid

Simulated site · MG-DEMO-01

Performance · fuel reconciliation and uncertainty

Overview | Foundation | Health | **Performance** | Findings | Work | Financials | Evidence

Service | Dispatch | **Fuel** | Renewable | Backup

Fuel balance: evidence limited — Illustrates the minimum balance terms; no residual conclusion yet

Opening evidence	Recorded deliveries	Expected use	Residual
200 L	Unknown	4.9–5.9 L	Unavailable
Reported tank level at selected start	Absence of record does not prove none occurred	Modelled: 18 kW for 60 min × 0.30 L/kWh ± 10%	Closing corroboration and delivery completeness needed

Fuel balance at a glance

Opening + deliveries − use − closing

Opening tank	200 L
+ Deliveries	Unknown
− Generator use	4.9–5.9 L
− Closing tank	Unknown
= Unexplained	Cannot compute

Hatched bars are missing evidence. A residual is shown only when every term has a supported range.

Balance bridge

TERM	EVIDENCE BASIS	TREATMENT
Opening inventory	Tank level reading	Retain sensor uncertainty
Plus deliveries	Delivery / operator records	Missing completeness blocks precise reconciliation
Less generator consumption	Measured output × configured coefficient	Modelled range, not fuel-meter reading
Less closing inventory	Tank reading / witnessed dip	Require a supported closing bound
Unexplained residual	Opening + delivery − use − closing	Compare with combined uncertainty; never infer theft

ⓘ A fuel residual is unexplained inventory movement with uncertainty. It cannot establish theft or be added automatically to dispatch-associated fuel exposure.

Inspect existing generator fuel interpretation

The dispatch Finding already has a bounded modelled fuel quantity for Candidate portions. This is a separate technical interpretation from a complete fuel-inventory balance.

Open dispatch consequence | Inspect fuel basis ↗

Canonical basis: v6.9 8.22 · **Question:** Can the fuel inventory movement be reconciled?
Preserved: Residual, uncertainty and expected-consumption basis stay explicit.
Adapted: Shows the missing-evidence state instead of inventing a fuel-loss story for Kobo.
Deferred: Fuel ERP and theft detection.
Content owners: FuelBalanceWindow · OperationalRecord · AcceptedEvidence · **Prototype route:** #/fuel

Sites › Kobo Mini-grid › Performance

Kobo Mini-grid

Simulated site · MG-DEMO-01

Performance · backup start and service recovery

Overview | Foundation | Health | **Performance** | Findings | Work | Financials | Evidence

Service | Dispatch | Fuel | Renewable | **Backup**

Backup readiness: Unknown Generator operation alone is not a witnessed start-and-recovery test

Outages in window	Start success	Last witnessed test	Critical service recovery
0	Unknown	No record	Unknown
No supply interruption seen	No outage to respond to	Test history not connected	Clinic need not measured

Backup event chains

Same chain as the cold-room Backup view: grid → start → generator → power → load → service

WHEN	GRID	START REQUESTED	GENERATOR	POWER TO LOAD	LOAD RESPONSE	OUTCOME FOR SERVICE	RESULT
No outage in the selected hour	—	—	—	—	—	—	No event

Backup readiness evidence

REQUIRED STEP	AVAILABLE EVIDENCE	CURRENT CLAIM
Supply interruption	No complete backup event in selected hour	No outage-chain assessment
Start command and state	Historical generator state reports	Operation observed; test not established
Power restored to critical service	Clinic delivered-power series	Requested-service baseline missing
Witnessed outcome	No accepted test record	Readiness remains unverified

Operating evidence

The generator is observed running in the selected dispatch period. This supports runtime reconstruction, not a general claim that automatic backup will succeed on demand.

Open Dispatch →

Next evidence

Run a witnessed start-to-service test and retain command, generator state, power-path and critical-service observations.

Open Actions →

Canonical basis: v6.9 8.24 · **Question:** What does the evidence say about backup readiness?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: BackupEventChain · ServiceWindow · CapabilityReadiness · **Prototype route:** #/backup

Kobo Mini-grid

Simulated site · MG-DEMO-01

Site-scoped Findings · operational conclusions and evidence limits

All | Cost | Risks | Opportunities

Scoped to Kobo · 3 open Same records, states and lenses as the global Findings list · no separate counts here Manage in Findings →

Findings at this site

Domain links back to the Performance view that produced the finding

FINDING	DOMAIN	PRIORITY	CONFIDENCE	STATE	MAIN CONSEQUENCE	LENS
F-104 · Generator support review · Supply cost per kWh Seen on 24 of the last 30 days	Dispatch	High	Low	Open	$3.31–3.97 cost exposure · selected hour	Cost
F-108 · Battery output-limit report missing · Evidence quality	Evidence	Medium	High	Open	30 min cannot be classified	Risks
F-109 · Productive-use headroom · Productive electricity sales Spare on 3 of 5 well-observed days	Renewable	Medium	Medium	Recommendation	18–20 kW usable · separate 5-day study	Opportunities

Early warnings · not priced yet

Early warning **Battery below its reserve**
Kobo · Battery and generator life · 5 of the last 30 days · lowest 22%

Early warning **Generator service due soon**
Kobo · Battery and generator life · 462 of 500 running hours

Opening a finding moves you to the Findings area; the breadcrumb keeps this site. This list has no status model of its own.

Canonical basis: v6.9 8.25 · **Question:** Which Findings belong to this site?
Preserved: Scoped index only; removing this tab loses a filter, not a distinct object.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: FindingIndex · **Prototype route:** #/sitefindings

Sites › Kobo Mini-grid › Work

Kobo Mini-grid

Simulated site · MG-DEMO-01

Site-scoped work · the same managed records as global queues

Overview | Foundation | Health | Performance | Findings | **Work** | Financials | Evidence

Actions (2) | Incidents (0) | Maintenance (0)

Scoped to Kobo · 2 actions Same records and states as the global Actions list — Manage in Actions →

Actions at this site

Due dates are not yet set in this example

ACTION	SOURCE	OWNER	DUE	STATE	SUCCESS CRITERION
ACT-048 · Review generator stop setting	F-104	Site controls lead	Not set	Verification window	Less generator running time with no service or reserve trade-off
ACT-049 · Restore battery-limit reporting	F-108	Instrumentation lead	Not set	Open	Battery output limit reported every 15 minutes

Opening a record moves you to its own area; the breadcrumb keeps this site. New work starts from the global create flow with this site filled in.

Canonical basis: v6.9 8.26 · **Question:** What work is active at this site?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: ActionIndex · IncidentIndex · MaintenanceIndex · **Prototype route:** #/sitework

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Demo workspace
3 sites · illustrative records

Bineyame Operations / Demo estate

Find a screen
Window: Event window
BY

Assets ›

Assets

Physical asset inventory · site context, condition and work

Inventory | Warranty & vendors | Register asset

All assets | Needs attention | Work due | Evidence limited | Warranty attention

7 registered · 3 sites Condition and evidence quality are separate columns and never merged

+ Register asset

Needs attention
3
Condition or evidence · BAT-001, GEN-CR-01, CHG-002

Work due
5
Open actions, repairs and scheduled maintenance

Warranty attention
4
1 at risk · 1 claimable · 2 not recorded
View detail →

Asset inventory

Opening an asset moves you to its own pages; the site stays in the breadcrumb

ASSET	TYPE	SITE	CONDITION	EVIDENCE	OPEN FINDINGS	OPEN WORK	WARRANTY
BAT-001 · Kobo battery	Battery	**Kobo**	Watch	Limited	0	1 · PM-0447	At risk to 1 Mar 2034
GEN-001 · Kobo generator	Generator	**Kobo**	Unknown	Running time only	1 · F-104	2 · ACT-048, PM-0441	Expired to 15 Jan 2026
INV-001 · Kobo solar inverter	Solar inverter	**Kobo**	Unknown	Output only	0	0	Not recorded
CMP-003 · Cold-room compressor	Compressor	**GreenHarvest**	Unknown	Context not comparable	0	1 · PM-0450	Active to 30 Jun 2028
GEN-CR-01 · Backup generator	Generator	**GreenHarvest**	Needs attention	Failed-start evidence	1 · F-118	2 · WR-019, PM-0455	Expired to 31 Mar 2026
ATS-CR-01 · Transfer switch	Transfer switch	**GreenHarvest**	Unknown	None · not mapped	0	0	Not recorded
CHG-002 · Connector 2 charger	DC charger	**Depot**	Needs attention	Interruption observed	1 · F-130	2 · ACT-061, WR-021	Claimable to 31 Aug 2028

ⓘ Detailed asset pages are shown for BAT-001 in this mock-up. Other assets open from their site pages.

Canonical basis: v6.9 8.3 · **Question:** Which physical assets merit a closer look?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: AssetDirectorySummary · AssetHealthSummary · **Prototype route:** #/assets

Findings ›

Findings

Evidence-backed operating conclusions · one global object model

All findings | Cost | Risks | Opportunities

Open findings	High priority	Low confidence	Awaiting verification
6	2	1	0
Across 3 sites	Decision needed	Evidence constrained · see the chip	Action taken; outcome open

All states | Open | Recommendation

Findings across the estate

Lens is how a finding is read, not a second object · state is a filter

FINDING	SITE	PRIORITY	CONFIDENCE	STATE	MAIN CONSEQUENCE	LENS
F-104 · Generator support review · Supply cost per kWh Seen on 24 of the last 30 days	Kobo Mini-grid	High	Low ⓘ	Open	$3.31–3.97 cost exposure · selected hour	Cost
F-108 · Battery output-limit report missing · Evidence quality	Kobo Mini-grid	Medium	High	Open	30 min cannot be classified	Risks
F-109 · Productive-use headroom · Productive electricity sales Spare on 3 of 5 well-observed days	Kobo Mini-grid	Medium	Medium ⓘ	Recommendation	18–20 kW usable · separate 5-day study	Opportunities
F-118 · Backup response failed · Cooling continuity · Stock kept saleable 3 grid losses in 30 days · backup started by itself on 2	GreenHarvest Cold Room	High	High	Open	Product conditions exposed · not priced	Risks
F-119 · Temperature excursion · Stock kept saleable	GreenHarvest Cold Room	Medium	High	Open	25–35 min above 4°C · peak 8.2°C	Risks
F-130 · Charging session interrupted · Buses ready to run Connector 2 status also unknown on 15 Sep	Depot Charging Hub	Medium	Medium ⓘ	Open	1 interrupted session · lost revenue unknown	Risks

Early warnings · not priced yet

Risk **Repeated maintenance without a result**
GreenHarvest · Cooling continuity · Backup generator visited 3 times in 60 days (14 Aug, 18 Sep, 6 Oct); the automatic start still failed

Early warning **Battery below its reserve**
Kobo · Battery and generator life · 5 of the last 30 days · lowest 22%

Early warning **Backup start test overdue**
GreenHarvest · Compressor life · Monthly test due 2 Oct, not done

Early warning **Charger fault and overdue inspection**
Depot · Charger life · Connector 2 fault on 6 Oct · inspection overdue since 30 Sep

Risk **Connector 2 out of service**
Depot · Buses ready to run · 1 h out and 1 h unknown on 6 Oct

Risk **Peak close to the import limit**
Depot · Room before expansion · 76 kW against 80 kW

Early warning **Generator service due soon**
Kobo · Battery and generator life · 462 of 500 running hours

Opportunity **Chargers mostly idle**
Depot · Charger use · Under a fifth of daily capacity used

ⓘ Priority, confidence and workflow state stay separate. Opening a finding moves you to Findings with its site in the breadcrumb.

Canonical basis: v6.9 8.4 · **Question:** Which supported conclusions deserve action?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: FindingIndex · FinancialConsequence · ActionSummary · **Prototype route:** #/findings

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3 sites · illustrative records

Bineyame Operations / Demo estate

Find a screen
Window: Event window
BY

Actions

Actions

Interventions, owners and declared outcome criteria

Open actions	Due / overdue	Verification window	Verified in 30 days
4	Not set	1	0
Accepted work	Due dates not yet set in this example	ACT-048	No closed verification yet

Managed actions

Lifecycle: Open → In progress → Completed → Verification window → Verified, Ineffective or Inconclusive

ACTION	SITE	SOURCE	OWNER	STATE	SUCCESS CRITERION
ACT-048 · Review generator stop setting	**Kobo**	**F-104**	Site controls lead	Verification window	≥25% less avoidable generator time; no reserve or service trade-off
ACT-049 · Restore battery-limit reporting	**Kobo**	**F-108**	Instrumentation lead	Open	Output limit reported every 15 minutes
ACT-051 · Witness automatic-start test	**GreenHarvest**	**F-118 · INC-203**	Maintenance lead	Open	Start requested → generator runs → power → cooling
ACT-061 · Investigate connector 2	**Depot**	**F-130**	Charging operations lead	Open	Complete witnessed session; no repeat fault

ⓘ Verification is a separate, evidence-based outcome. This mockup does not issue commands to real equipment.

Canonical basis: v6.9 8.5 · **Question:** Who owns the next intervention and what will prove it worked?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: ActionIndex · VerificationSummary · **Prototype route:** #/actions

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Find a screen
Window: Event window
BY

Incidents

Incidents

Operational event cases · linked Findings and work

Open incidents	High severity	Awaiting evidence	Closed verified
1	1	1	0
INC-203	Cold-room backup failure	Closure needs a witnessed start	In the selected window

Incident cases

States: New → Acknowledged → Investigating → Work open → Awaiting evidence → Closed verified

INCIDENT	SITE	START	SEVERITY	STATE	OWNER	FINDINGS / WORK
INC-203 · Backup response failure	GreenHarvest	6 Oct 19:11	High	Awaiting evidence	Maintenance lead	F-118 · F-119 / WR-019 · ACT-051

Case boundaries

A correlated power-to-cooling chain supports this incident. Arbitrary data gaps or unrelated observations are not promoted into cases.

Canonical basis: v6.9 8.6 · **Question:** Which operational cases need coordinated investigation?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: IncidentIndex · **Prototype route:** #/incidents

Maintenance

Physical maintenance work and outcome verification

Corrective | Preventive · scheduled

Open work
2
WR-019, WR-021

Verification overdue
1
WR-019 · witnessed start test missing

Preventive due in 30 days
4
See Preventive
View detail →

Repeat visits without a result
1
Backup generator · 3 visits, start still failing
View detail →

Corrective work requests

Completed is activity evidence only; outcome is verified separately

WORK REQUEST	ASSET / SITE	OWNER	STATE	VERIFICATION	WARRANTY
WR-011 · Starter battery replaced · 14 Aug	GEN-CR-01 · GreenHarvest	Maintenance lead	Completed	Not checked	Expired
WR-016 · Control relay reset · 18 Sep	GEN-CR-01 · GreenHarvest	Maintenance lead	Completed	Not checked	Expired
WR-019 · Starter and control inspection	GEN-CR-01 · GreenHarvest	Maintenance lead	Completed	Pending · witness test missing	Expired
WR-021 · Connector 2 repair	CHG-002 · Depot	Charging operations lead	Open · waiting for diagnostics	Required	Claimable

ⓘ Work requests share the managed-task lifecycle with Actions: Open → In progress → Completed → Verification window → Verified, Ineffective or Inconclusive.

Canonical basis: v6.9 8.7 · **Question:** Which physical work is complete, and which outcomes remain unverified?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: MaintenanceIndex · VerificationSummary · **Prototype route:** #/maintenance

Evidence

Portfolio evidence

Cross-site claim readiness, recurring gaps and instrumentation priorities

Sites with limited claims
3
All three domains have explicit gaps

Priority evidence actions
3
PCS, loading context and session completeness

Unsupported claims
Excluded
No invented loss, spoilage or RUL values

Capability coverage across sites

Each row links to site-owned readiness; this page does not independently score the claims

SITE / DOMAIN	OPERATIONAL CAPABILITY	READINESS	CURRENT CEILING	NEXT EVIDENCE
Kobo · mini-grid	Generator support	Limited	Candidate in supported portions	Restore 14:50 PCS capability
GreenHarvest · cold-chain	Temperature / backup	Limited	Excursion and failed-start chain	Loading context + witnessed recovery test
Depot · EV charging	Session / availability	Limited	Observed outcomes; unknown time bounded	Missing session end and connector status

Recurring evidence gaps

GAP	SITES	UNLOCKS	NEXT STEP
Device stops reporting (battery controller, connector 2)	Kobo · Depot	Dispatch classification · availability	**ACT-049, ACT-061 →**
Points wired but not mapped (transfer switch, battery temperature)	GreenHarvest · Kobo	Backup chain · battery condition and warranty	Map transfer switch →
Manual records not taken (closing fuel dip, stock in chamber)	Kobo · GreenHarvest	Fuel balance · product exposure	**Operational records →**
Tests never done (battery capacity, witnessed start)	Kobo · GreenHarvest	Capacity · backup readiness	**Preventive plan →**

ⓘ Source connectivity is not physical asset health. Evidence Full/Limited/None/N-A is a separate vocabulary from Healthy/Watch/Degraded.

Canonical basis: v6.9 8.8 · 8.35 · **Question:** Where do evidence gaps constrain estate-wide decisions?
Preserved: Global roll-up links down to site-owned readiness.
Adapted: Compact domain-capability matrix for three sites.
Deferred: A duplicate global raw-record browser.
Content owners: PortfolioEvidenceMatrix · CapabilityReadiness · **Prototype route:** #/globalevidence

Financials

Portfolio financials

Estate consequences with source-site ownership and bounded interpretation

Site-owned consequences — Values grouped by basis and period; no financial assessment

Consequence register

No independent repricing or technical re-derivation

SITE	CATEGORY	WINDOW / QUANTITY	VALUE	BASIS / LIMIT	SOURCE
Kobo	Candidate operating-cost exposure	6 Oct · 14:20–15:20	$3.31–3.97	Modelled fuel + scenario accrual	Site bridge →
GreenHarvest	Potential product exposure	6 Oct · backup excursion	Unpriced	Stock / product loss not established	Site evidence →
Depot	Gross tariff estimate	6 Oct · 08:00–16:00 · 120 kWh	$36.00	Scenario tariff; not collections	Site bridge →

ⓘ These values are not an additive profit or savings total. Categories, time periods, uncertainty and overlapping technical bases differ.

Aggregation rule

Aggregate distinct eligible consequence-line IDs by category, currency and compatible window. Reuse the exact site consequence revision; preserve exclusions and assumption references.

Canonical basis: v6.9 8.9 · 8.36 · **Question:** What consequences are supported across the estate?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: PortfolioConsequenceRollup · FinancialConsequenceRevision · **Prototype route:** #/globalfinancials

Kobo Mini-grid

Simulated site · MG-DEMO-01

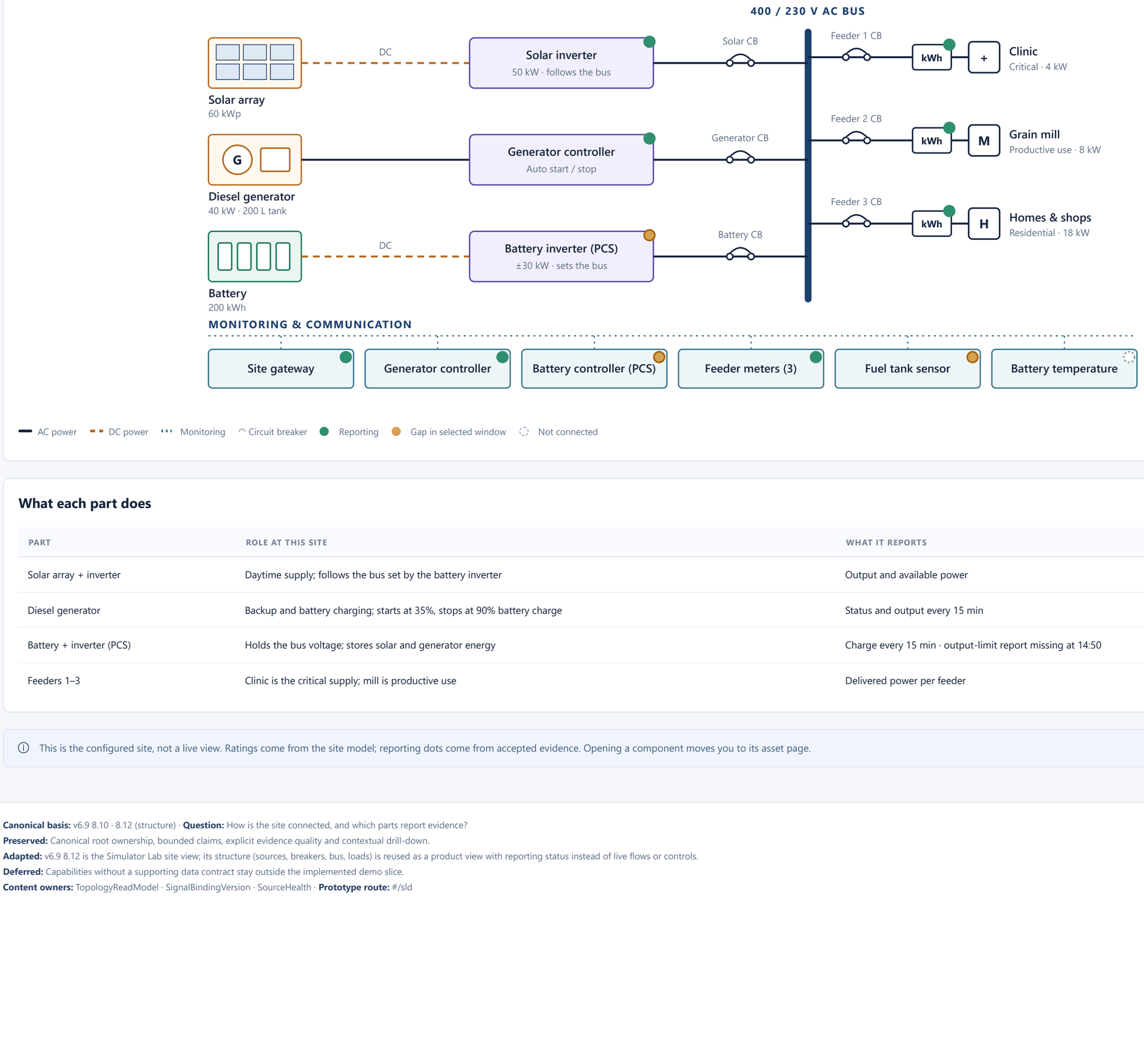
Foundation · single line diagram

Configured site · effective 6 Oct 2026 — Single line diagram · what is connected to what, and which parts report evidence

Site evidence →

Kobo Mini-grid · AC-coupled mini-grid

Configured equipment, protection and metering · reporting status for the event window, not live power flow



What each part does

PART	ROLE AT THIS SITE	WHAT IT REPORTS
Solar array + inverter	Daytime supply; follows the bus set by the battery inverter	Output and available power
Diesel generator	Backup and battery charging; starts at 35%, stops at 90% battery charge	Status and output every 15 min
Battery + inverter (PCS)	Holds the bus voltage; stores solar and generator energy	Charge every 15 min · output-limit report missing at 14:50
Feeders 1–3	Clinic is the critical supply; mill is productive use	Delivered power per feeder

ⓘ This is the configured site, not a live view. Ratings come from the site model; reporting dots come from accepted evidence. Opening a component moves you to its asset page.

Canonical basis: v6.9 8.10 · 8.12 (structure) · **Question:** How is the site connected, and which parts report evidence?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: v6.9 8.12 is the Simulator Lab site view; its structure (sources, breakers, bus, loads) is reused as a product view with reporting status instead of live flows or controls.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: TopologyReadModel · SignalBindingVersion · SourceHealth · **Prototype route:** #/sld

Operational records

Operational records

Operator logs, manual readings, shift notes and site visits

Records in window
6
Logs, readings, notes and visits

Feeding evidence
6
Labelled Operator record where used

In review
1
REC-321

Requested, not yet done
2
Closing fuel dip · stock in chamber

All | Operator log | Manual reading | Shift note | Site visit

Operational records

What people on site recorded, and the evidence each one feeds

RECORD	TYPE	SITE	WHEN	BY	WHAT IT SAYS	FEEDS	STATE
REC-305	Manual reading	Kobo	6 Oct 14:15	Site operator	Fuel tank dip: 198 L (sensor read 200 L)	Fuel balance · opening →	Accepted
REC-306	Shift note	Kobo	6 Oct 15:30	Shift supervisor	Generator left running to charge the battery; stop setting 90% unchanged	F-104 context →	Accepted
REC-307	Manual reading	Kobo	Requested	—	Closing fuel tank dip not yet taken	Fuel balance · closing →	Requested
REC-311	Operator log	GreenHarvest	6 Oct 19:14	Night operator	Generator started by hand after no automatic start	INC-203 timeline →	Accepted
REC-312	Site visit	GreenHarvest	6 Oct 21:40	Technician	Starter and control circuit inspected; connection re-made	WR-019 record →	Accepted
REC-313	Manual reading	GreenHarvest	Requested	—	Stock in chamber not recorded	Product exposure →	Requested
REC-320	Shift note	Depot	6 Oct 14:20	Depot attendant	Driver reported connector 2 stopped mid-charge; vehicle moved to connector 1	F-130 context →	Accepted
REC-321	Site visit	Depot	6 Oct 16:30	Charging technician	Connector 2 checked visually; no damage; diagnostics not exported	ACT-061 →	In review

Record a manual reading

Fills REC-307 when submitted

Site
Kobo Mini-grid

Reading
Fuel tank dip

Value
e.g. 192 L

Taken at
6 Oct 2026 · 16:00 EAT

Submit for review

How records are used

Records are kept apart from measured readings and are labelled **Operator record** wherever they support a conclusion.

A requested record that is never taken stays a visible gap; it is not filled in.

Canonical basis: v6.9 Owner outline · 8.33 basis (Operator record) · **Question:** What did people on site record, and what does it feed?
Preserved: Canonical root ownership, bounded claims, explicit evidence quality and contextual drill-down.
Adapted: Populated example data and a concise operator-first hierarchy. Technical detail remains available through evidence and analysis.
Deferred: Capabilities without a supporting data contract stay outside the implemented demo slice.
Content owners: OperationalRecord · **Prototype route:** #/oprecords